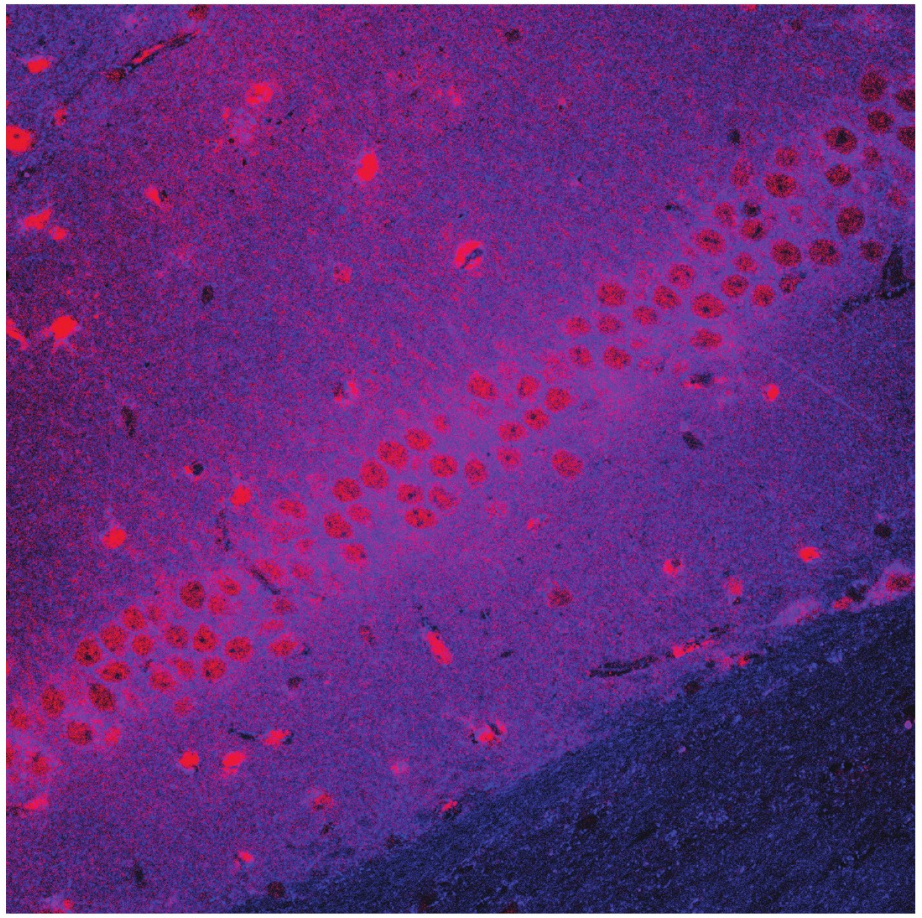


Oleg Senkov/Shutterstock

Neurons in the mouse hippocampus, an area involved in affective disorders

Carlson and Birkett (2022), Chapter 17 opener, p.548. Image: Oleg Senkov/Shutterstock.

BEHAVIORAL NEUROSCIENCE 392

Schizophrenia and the Affective Disorders

Disrupted Thought and Mood: Circuits, Symptoms, and Treatment

Fall 2026

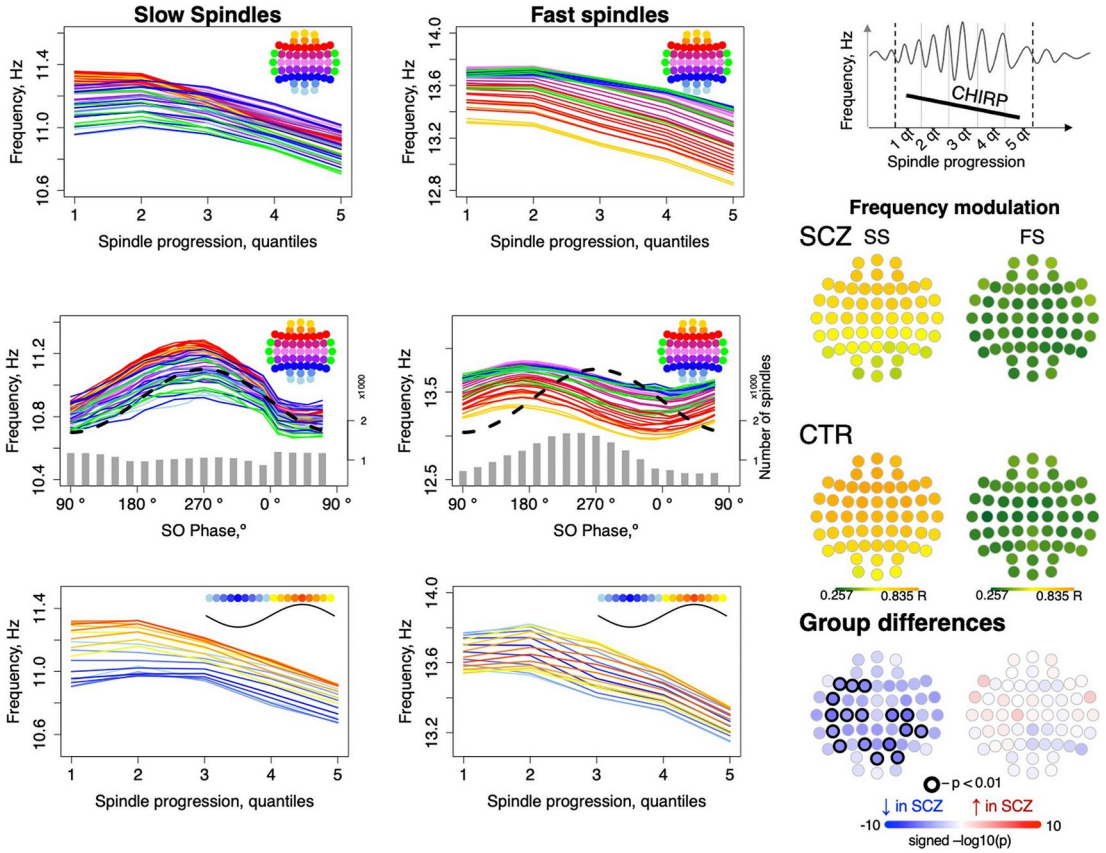
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Schizophrenia affects several dimensions of experience

Schizophrenia is a psychiatric disorder involving changes in perception, thought, motivation, and cognition. A diagnosis describes a pattern over time, including impairment and the exclusion of other explanations. It does not mean that a person has multiple personalities.

Positive symptoms are experiences added to ordinary functioning, such as hallucinations or delusions. Negative symptoms involve reductions in functions such as motivation, speech, or emotional expression. Cognitive symptoms include difficulties with attention, working memory, and organizing behavior.

These dimensions can vary independently within the same person. Reducing hallucinations does not necessarily restore working memory or motivation. Clinical and experimental measurements therefore need to specify which function changed, rather than reporting every outcome as improvement in schizophrenia.



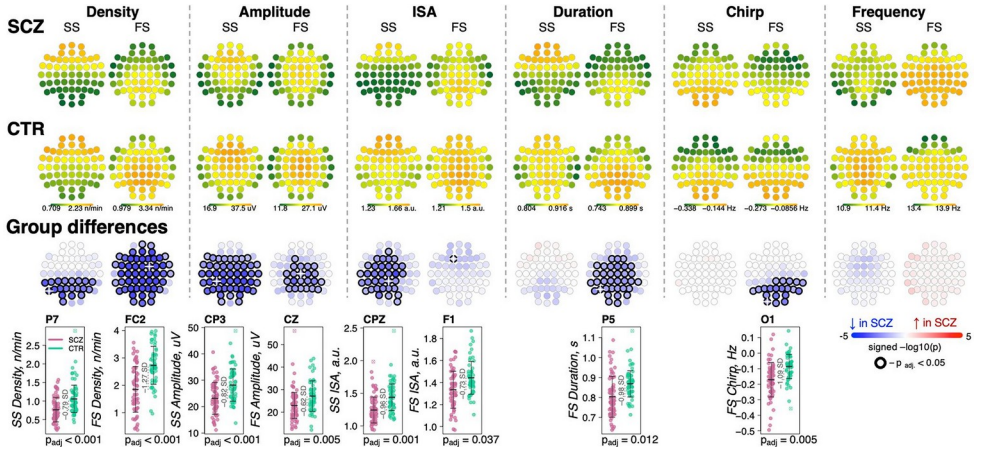
Kozhemiako et al. (2022), Fig. 4.

Sleep recordings reveal altered thalamocortical activity

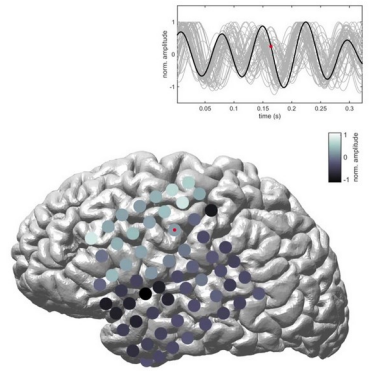
During non-rapid-eye-movement sleep, the cortex and thalamus generate organized electrical events. Sleep spindles are brief rhythmic bursts visible in an EEG. Researchers measured these events in people with schizophrenia and comparison participants, alongside waking auditory responses.

The schizophrenia group had reduced spindle density, meaning fewer spindles within the measured sleep period. The study also examined spindle properties and their timing relative to slower oscillations. These measurements describe organization within a sleeping network rather than a person's conscious report.

Spindles also have a spatial organization. A separate human cortical-electrode recording shows activity rotating across the sampled surface over several spindle cycles. Each electrode supplies its own voltage measurement, revealing a traveling pattern rather than every cortical location reaching its peak at the same instant.



Kozhemiako et al. (2022), Fig. 2.



Human spindle activity rotating across cortex. Muller et al. (2016).

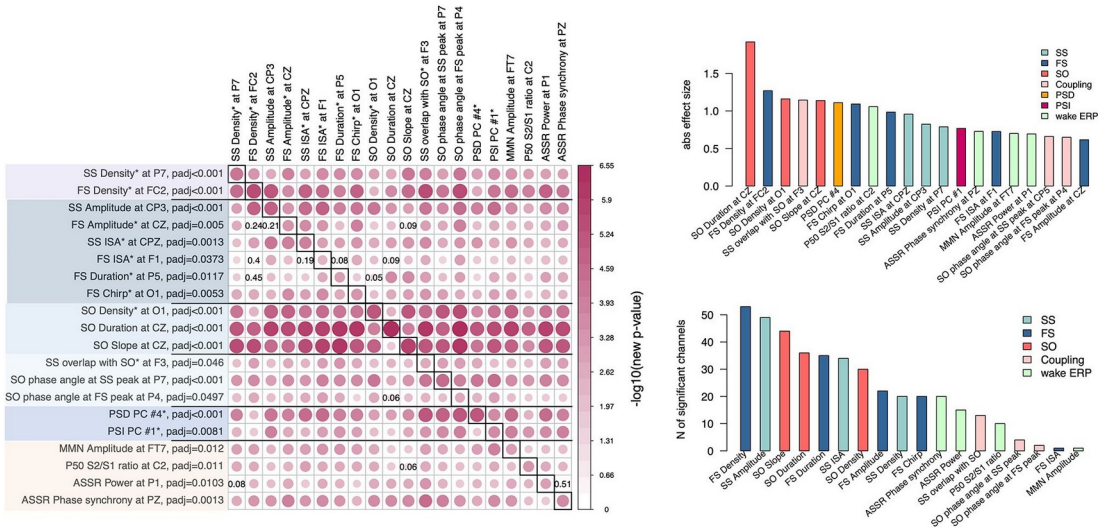
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Hallucinations differ from voluntary imagination

A hallucination is a perception-like experience without the corresponding external stimulus. Hearing a voice can feel perceptually real to the person experiencing it. It differs from deliberately imagining a voice, which usually remains recognized as an internally generated act.

A delusion is a strongly held belief that persists despite contrary evidence and is understood in its cultural and clinical context. Its content varies. The biological question concerns how perception, salience, inference, and learning produce or maintain the experience.

Psychotic symptoms can occur in several medical and psychiatric conditions, not only schizophrenia. Their presence therefore initiates a differential assessment. Substance effects, mood episodes, neurological illness, and the course of symptoms are relevant to identifying the underlying condition.



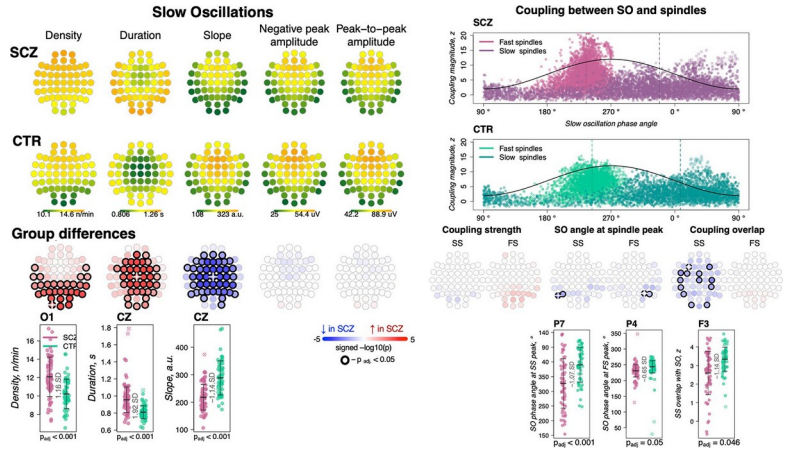
Kozhemiako et al. (2022), Fig. 7.

Slow oscillations and spindles lose coordinated timing

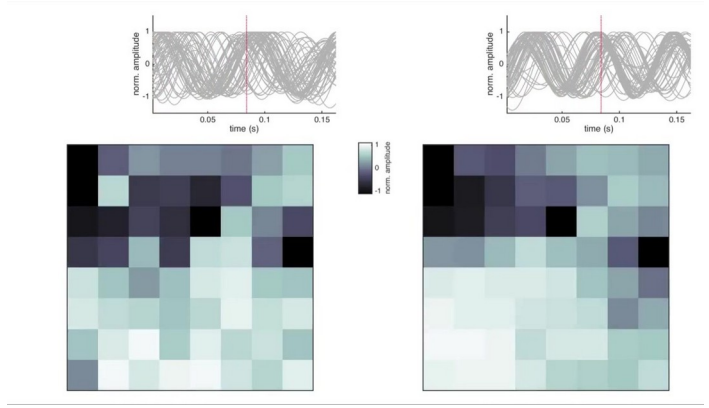
A slow cortical oscillation alternates between phases of lower and higher neuronal activity. Sleep spindles can occur preferentially during particular phases of this slower cycle. Their interaction organizes when thalamic and cortical activity occur together during sleep.

The schizophrenia study found an attenuated measure of this interaction. The result concerns temporal coordination, not simply a reduction in the amount of sleep. Fast and slow spindle properties also varied partly independently, indicating that one summary spindle measure could miss distinct changes.

A separate human sleep recording compares two spindle waves several minutes apart. Their activity rotates across the electrode array in similar spatial sequences. Repeated organization across the surface is a different timing measurement from the coupling between slow oscillations and spindles in the clinical study.



Kozhemiako et al. (2022), Fig. 3.



Two repeated spatial spindle patterns. Muller et al. (2016).

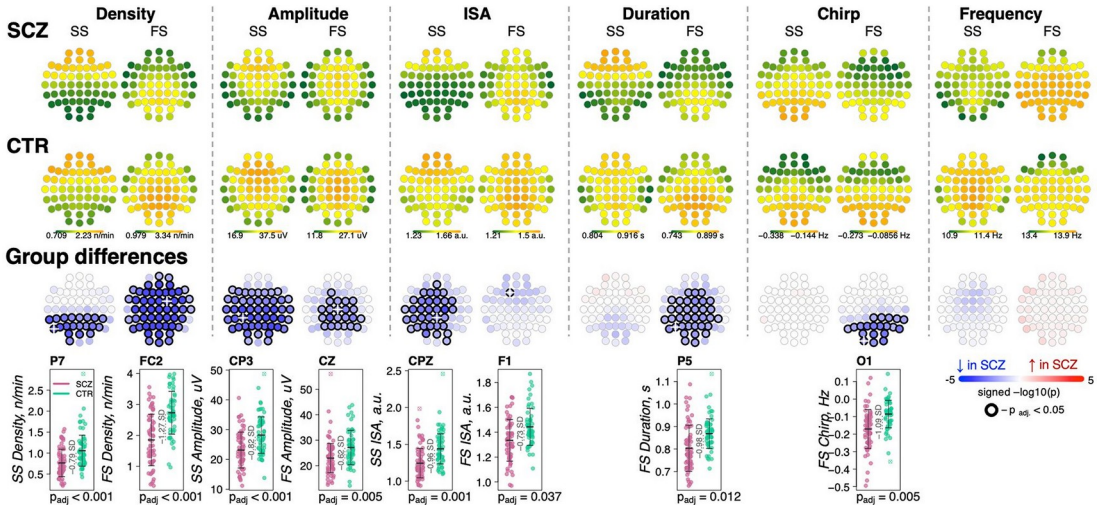
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Negative symptoms are not a measure of moral effort

Reduced initiation of activity is called avolition, while diminished pleasure is called anhedonia. These descriptions concern behavior and experience. They do not establish that a person is choosing to be inactive or failing to value other people.

Reduced outward emotional expression is also different from absence of internal feeling. A clinician observes facial movement, speech, and behavior, but must also ask about experience. The visible behavior and the reported subjective state supply different information.

Sleep disruption, depression, medication effects, and environmental circumstances can contribute to reduced activity. Assessing these factors helps distinguish primary negative symptoms from other causes. The distinction matters because the same observed behavior can arise through several biological and social routes.



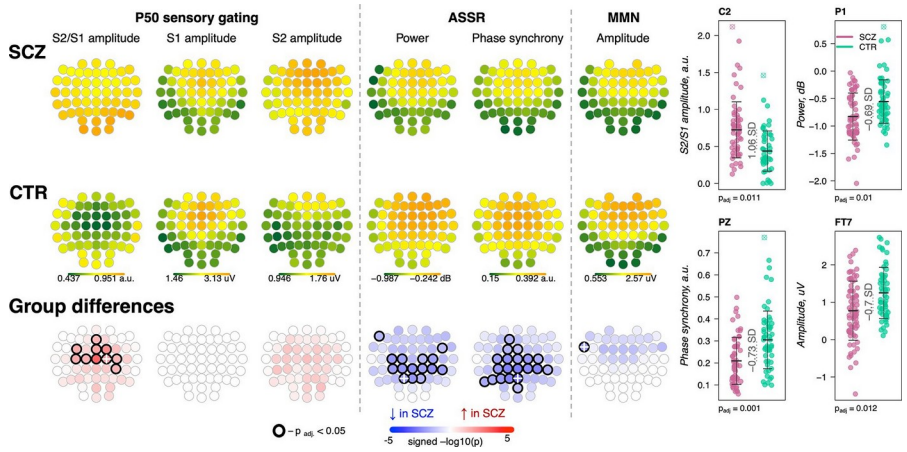
Kozhemiako et al. (2022), Fig. 2.

Waking and sleeping markers sample different processes

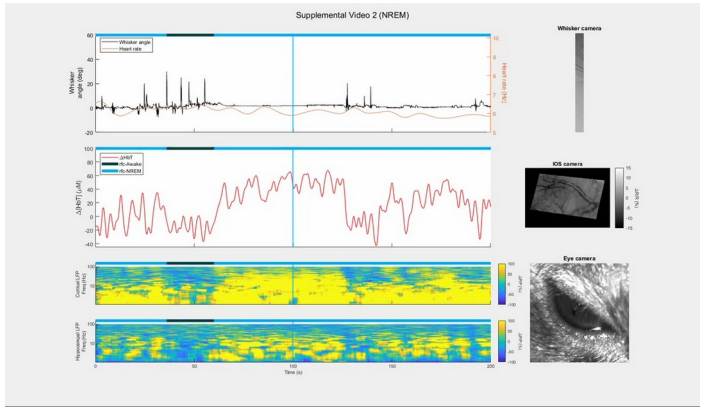
The investigators also measured auditory event-related potentials during waking. An event-related potential is an EEG response averaged around a defined stimulus. It samples how neural populations respond to that event, whereas spindle measurements sample spontaneous events during sleep.

Both waking and sleeping measurements differed in the schizophrenia group, but their abnormalities were not tightly coupled. A person with a change in one measure did not necessarily show a matching change in the other. The tests therefore sampled partly distinct physiological features.

Mouse sleep recordings can combine neural signals with visible behavior. During non-REM sleep, the filmed animal has little whisker movement despite an open eye, while cortical and blood-flow signals change. Eye opening alone therefore does not identify the physiological sleep state in this preparation.



Kozhemiako et al. (2022), Fig. 8.



Mouse non-REM sleep: behavior and physiology. Turner et al. (2020).

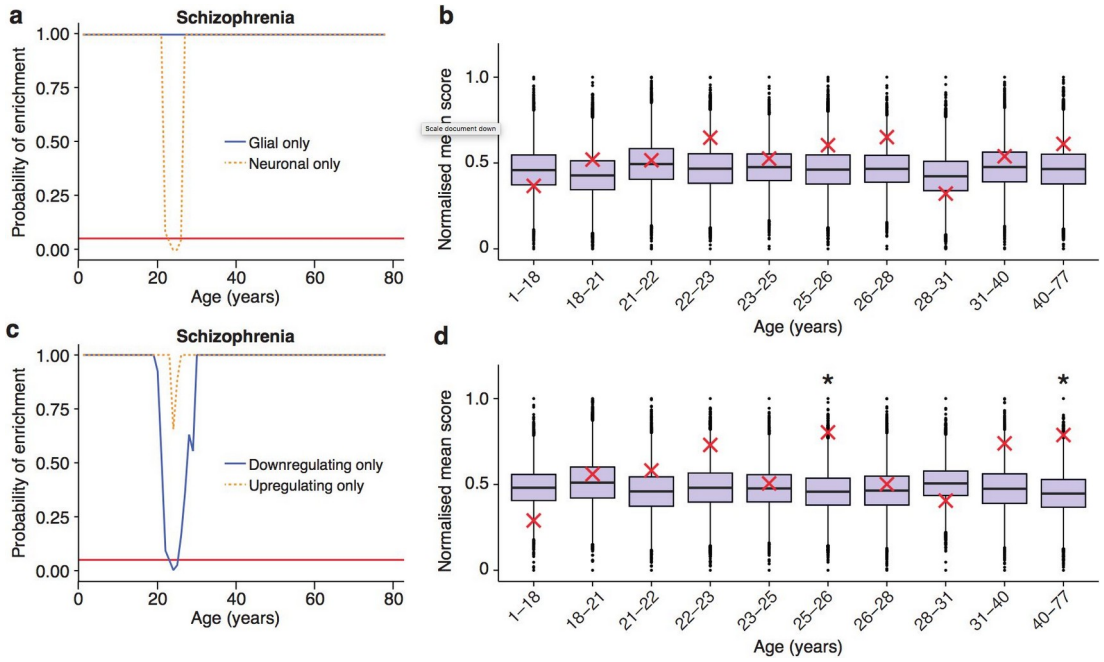
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Genetic risk changes probability, not destiny

Schizophrenia has a substantial genetic contribution, but most risk is distributed across many variants. A risk-associated variant changes probability within a population. It is different from a single mutation that reliably produces the same clinical syndrome in every carrier.

Relatives share genes and aspects of their environments. Twin and family comparisons help estimate inherited contributions, while molecular studies identify associated regions of the genome. Neither approach means that environment becomes irrelevant when genetic effects are measurable.

Genes affect proteins and regulatory processes within developing cells. Their effects can depend on cell type, developmental stage, and other genetic variation. A statistical association therefore begins a mechanistic question about when and where the relevant biology changes.



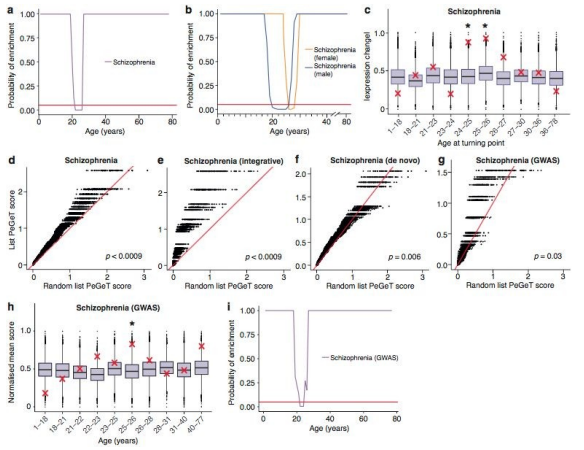
Skene et al. (2017), Fig. 6.

Gene-expression trajectories change across the lifespan

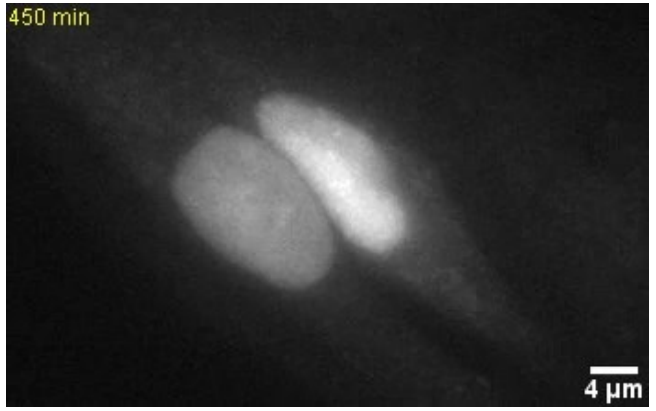
Researchers examined brain gene expression across ages in humans and mice. A transcriptome is the set of RNA transcripts measured in a sample. Tracking expression with age allows identification of groups of genes whose activity changes along related developmental trajectories.

The study described a period of especially large expression change during young adulthood. Synaptic genes and schizophrenia-associated gene sets participated in this reorganization. The finding concerned age-dependent gene regulation rather than a sudden appearance of a new chromosome or a single disease gene.

Gene activity also changes within individual living cells. A separate reporter experiment marks newly produced RNA with fluorescent protein, making active transcription sites visible. Cells receiving the same chemical stimulus show different time courses, rather than all producing RNA in a synchronized continuous stream.



Skene et al. (2017), Fig. 5.



Single-cell transcription dynamics. Larson et al. (2013).

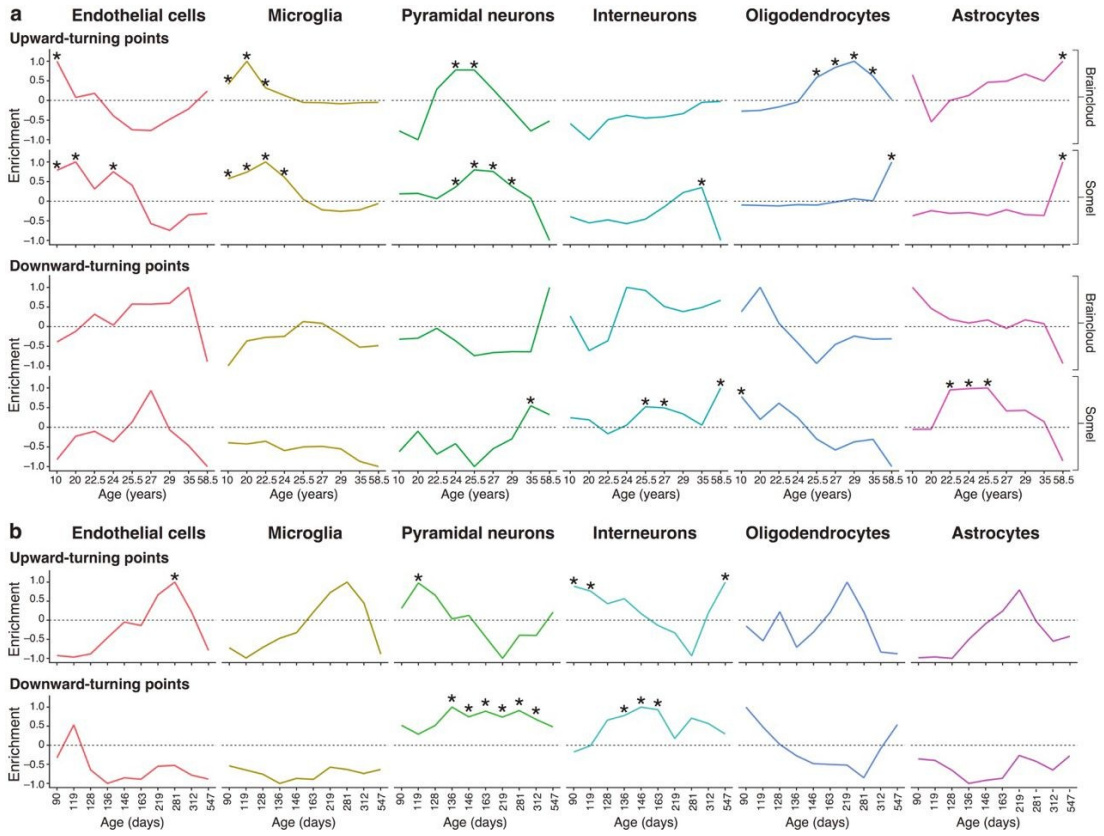
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Different cell types follow different expression schedules

A brain-tissue sample contains neurons, glia, vascular cells, and other cellular populations. Their RNA contributes to the measured transcriptome. The lifespan study identified age-related trajectories associated with several of these cell types rather than assigning every change to neurons alone.

Synaptic regulation changes as neural circuits mature, while glial and endothelial programs follow their own patterns. These processes can overlap in time without sharing the same cellular action. Their contribution to a bulk sample must therefore be interpreted with cell-type information.

The researchers compared human and mouse trajectories to examine conserved features. Similar timing patterns across species identify candidate biological programs, while differences in lifespan and developmental pace prevent a mouse age from being treated as the same calendar age in humans.



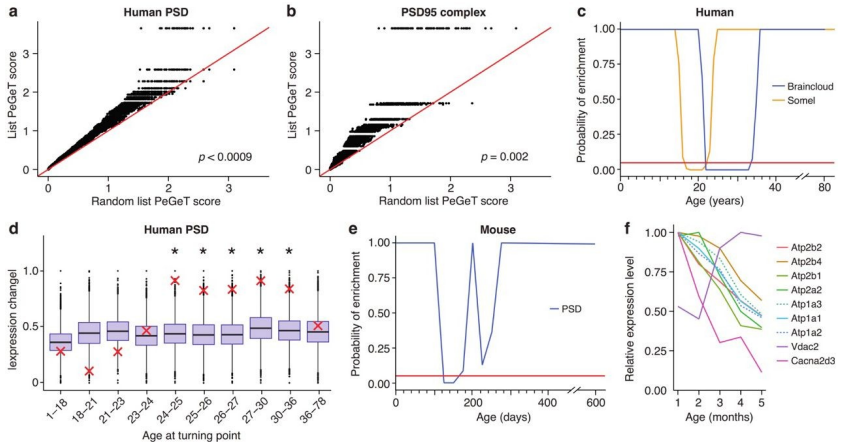
Skene et al. (2017), Fig. 3.

Developmental risk can precede the first clinical episode

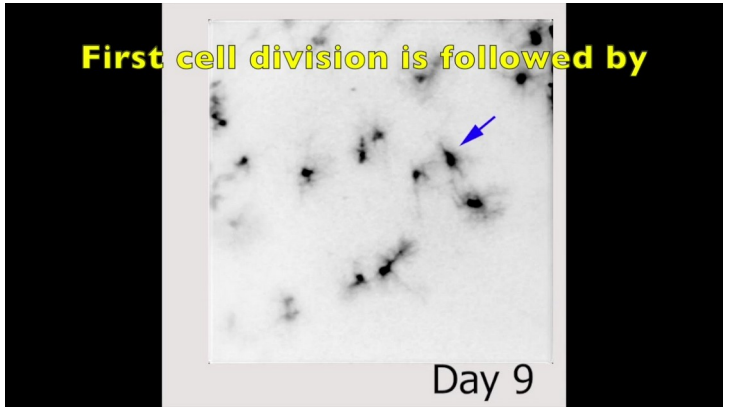
The first recognized psychotic episode may occur years after earlier developmental influences. Genes, prenatal conditions, and childhood experiences can affect circuits before symptoms become clinically apparent. The age at diagnosis is therefore not necessarily the age when every relevant biological change began.

Adolescence and young adulthood involve continuing changes in synapses, myelination, and cognitive control. These changes alter how established circuits process information. A vulnerability formed earlier can become more consequential when the demands and organization of the network change.

Development depends partly on how stem cells divide. Separate imaging of adult mouse hippocampus follows asymmetric neural stem-cell divisions across days, with daughters taking different cellular forms. These normal divisions demonstrate a specific developmental process rather than a recording of schizophrenia emerging in an animal.



Skene et al. (2017), Fig. 4.



Asymmetric neural stem-cell divisions. Guo et al. (2022).

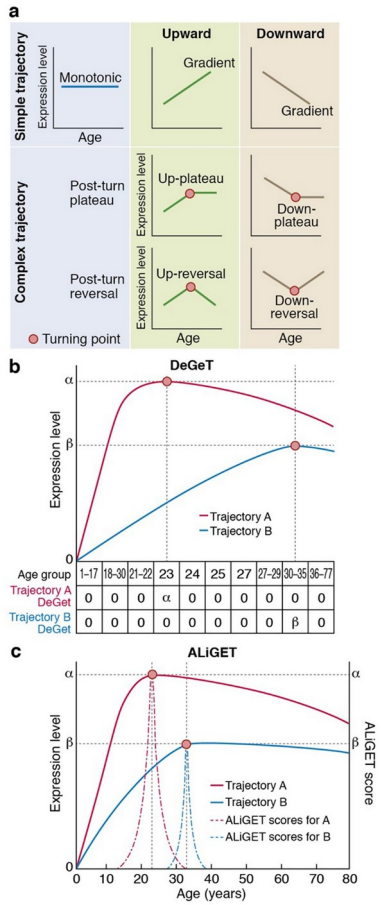
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Age-associated expression identifies a timing pattern

The lifespan analysis measured relationships between age and RNA expression. It did not experimentally advance or delay the developmental program in human participants. The strongest direct finding was an organized set of expression changes across the sampled lifespan.

The overlap with schizophrenia-associated genes suggested a biological intersection between developmental regulation and inherited risk. Such overlap can prioritize genes or pathways for further study. It does not identify which altered transcript is sufficient to produce a particular symptom.

Testing a causal mechanism would require additional interventions in suitable preparations. The descriptive trajectories instead establish when candidate molecular processes change, providing a temporal framework for experiments that target specific cells, genes, and developmental stages.



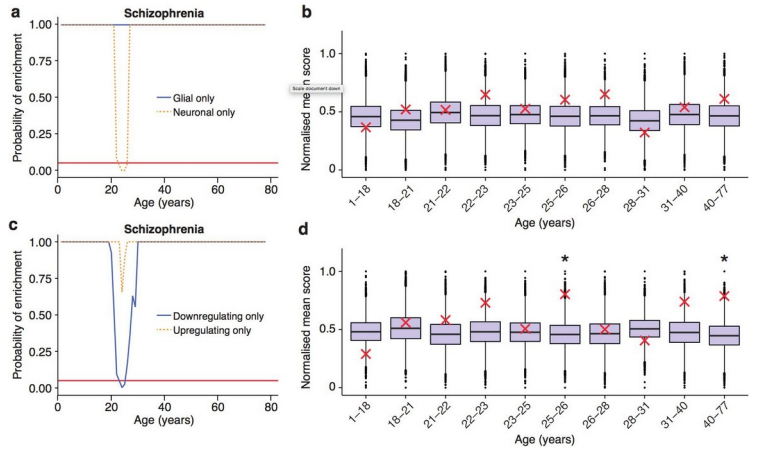
Skene et al. (2017), Fig. 1.

A shared diagnosis can contain different molecular routes

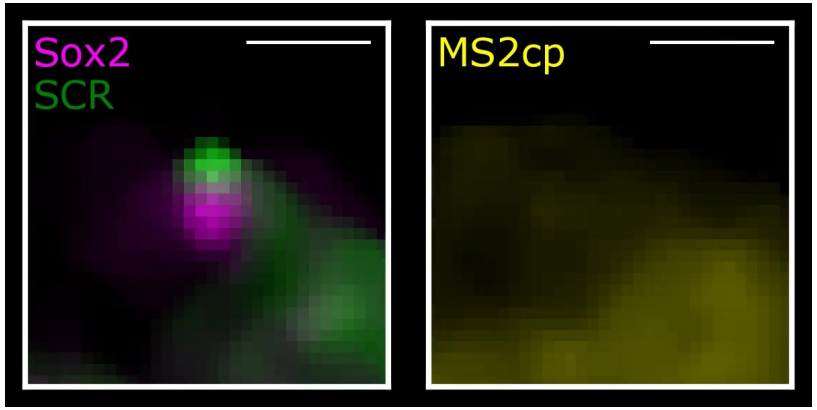
Many genes contribute small amounts of schizophrenia risk, and they influence several cellular systems. Their effects can converge on synaptic organization without being identical. Two people meeting the same diagnostic criteria may therefore differ in the molecular routes contributing to their symptoms.

The lifespan study examined coordinated gene sets rather than searching only for a single universally abnormal transcript. This approach detected changes in the organization of expression over age. The relevant biological unit was a changing program involving multiple genes.

Transcription can be examined at a single gene in living cells. Separate imaging labels the Sox2 gene region, a distant regulatory region, and newly made Sox2 RNA. Bursts of RNA production appear without sustained overlap of the two DNA labels in this preparation.



Skene et al. (2017), Fig. 6.



Sox2 transcription bursts and regulatory DNA. Alexander et al. (2019).

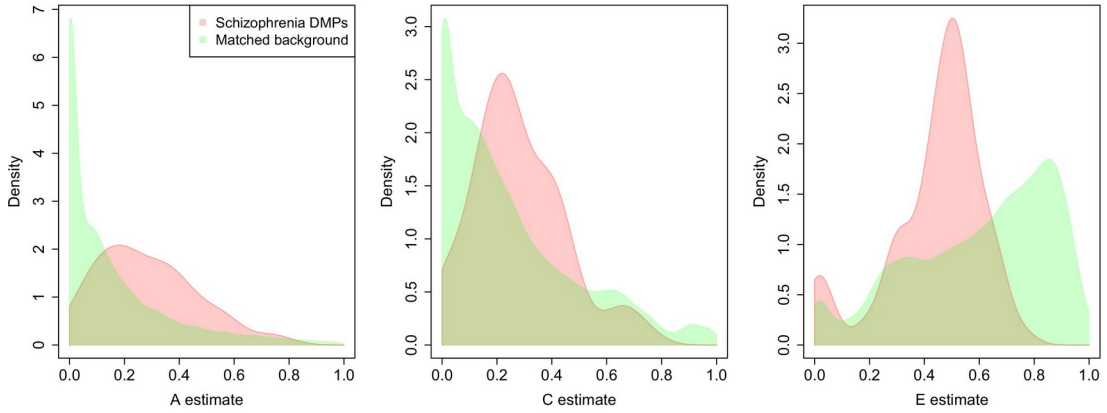
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Exposures leave different biological traces

Environmental contributions to schizophrenia include multiple exposures rather than one universal cause. Prenatal conditions, social adversity, and substance exposure can influence risk through different routes. A population association does not establish that a given exposure caused one person's disorder.

Epigenetic regulation changes how cells use DNA without changing its sequence. DNA methylation adds chemical marks at particular sites and can alter or reflect regulatory activity. The effects depend on the genomic location and the cell type carrying the mark.

Blood is easier to sample repeatedly than brain tissue, but its cells serve different functions. A methylation difference in blood can describe immune composition or exposure history. It cannot automatically be interpreted as the same change inside a cortical neuron.



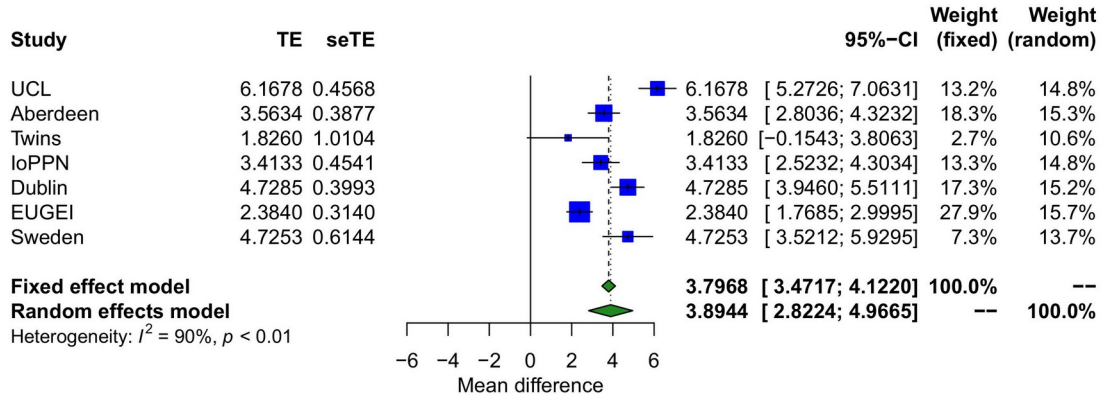
Hannon et al. (2021), Fig. 7.

Blood methylation records cell composition and exposure

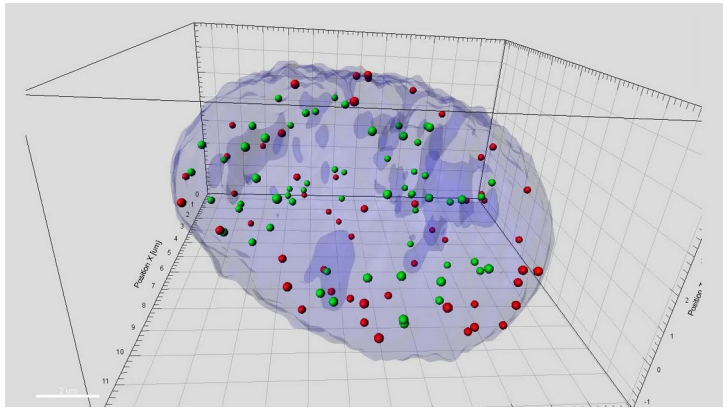
Researchers combined blood DNA-methylation data from several psychosis cohorts. They examined differences associated with psychosis, schizophrenia, and treatment-resistant schizophrenia. A meta-analysis combines results across datasets while accounting for features of the separate studies.

The affected groups differed in estimated blood-cell proportions and methylation signatures associated with smoking exposure. These features changed the interpretation of disease-associated sites because the sample's cellular mixture and exposure history contributed to the measured signal.

DNA is packaged with proteins inside the nucleus. Separate imaging of a mouse embryonic stem cell labels chromatin regions near the nuclear envelope and regions located farther inward. This spatial organization is a different measurement from chemical methyl marks measured on DNA in human blood.



Hannon et al. (2021), Fig. 3.



Chromatin organization in a stem-cell nucleus. Poleshko et al. (2019).

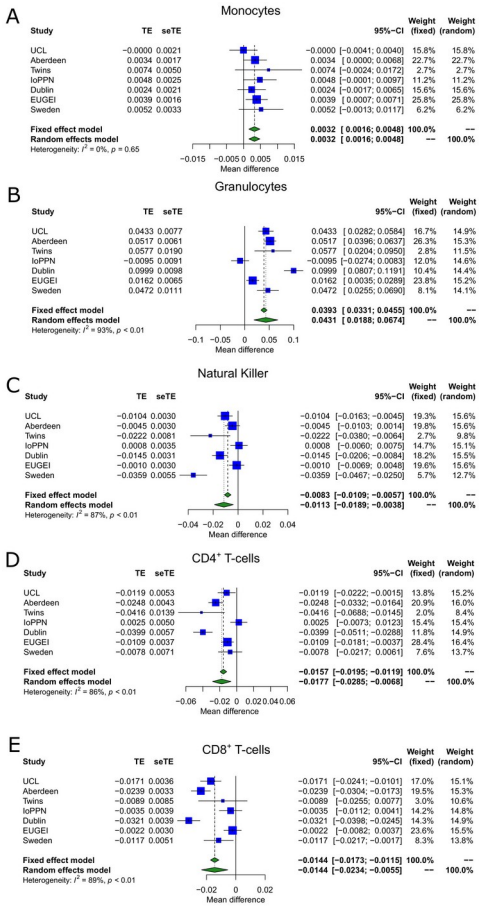
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A mixed-cell sample can change without every cell changing

Suppose a blood sample contains a larger proportion of one immune-cell population. Its methylation profile can shift because the sample contains different cells, even if methylation within each cell type remains unchanged. Cellular composition is therefore part of the measurement.

The psychosis study estimated these proportions and incorporated them into its analysis. This helped separate a difference in the mixture of cells from differences occurring within the sampled cellular populations. The two mechanisms can coexist in the same participant.

This distinction also applies to brain tissue containing neurons and glia. A molecular difference in a mixed sample needs an anatomical and cellular interpretation. Without it, a change in abundance can be mistaken for a universal change in every contributing cell.



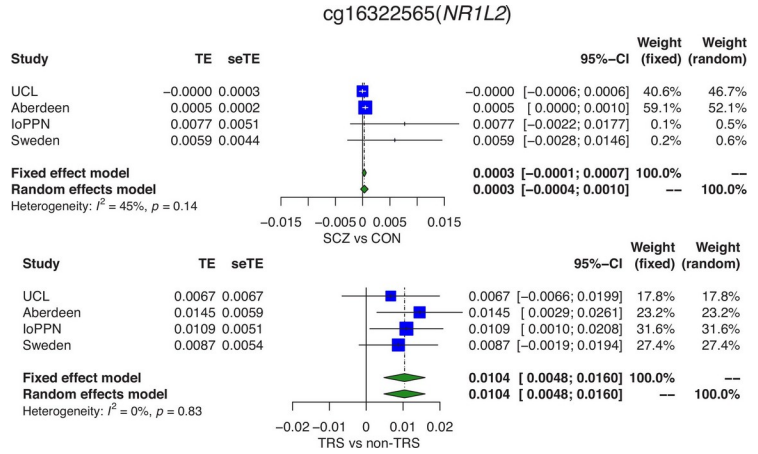
Hannon et al. (2021), Fig. 2.

Stress exposure does not specify diagnosis

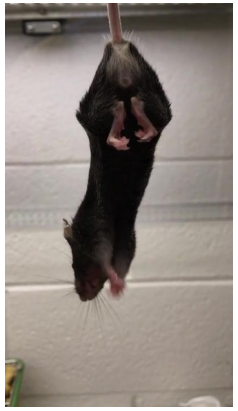
An exposure can increase risk without being necessary or sufficient for a disorder. Some exposed people do not develop schizophrenia, and some people with schizophrenia do not report the same exposure. This is expected when several pathways contribute to a heterogeneous outcome.

A biological mediator would lie between exposure and a later change in function. Demonstrating mediation requires more than observing that exposure and symptoms occur together. Researchers need evidence about timing and the specific process carrying the effect.

Manipulating a methylating enzyme can produce a defined experimental phenotype. A mouse lacking Dnmt3a in selected cells shows repeated forepaw movements in the recording. These stereotyped movements are the observed animal behavior; they are different from either a human psychosis diagnosis or a blood methylation signature.



Hannon et al. (2021), Fig. 8.



Forepaw stereotypies in a Dnmt3a-deficient mouse. Lavery et al. (2020).

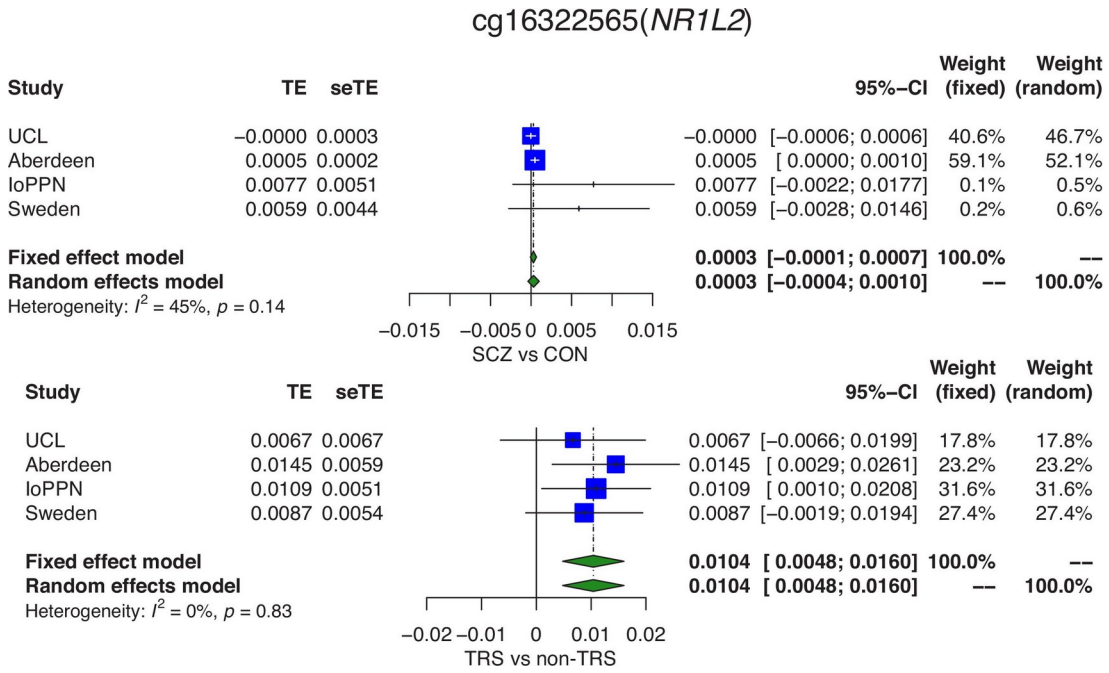
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Treatment-resistant groups show molecular differences

Treatment-resistant schizophrenia describes persistent symptoms despite adequate treatment attempts under clinical criteria. It is not simply a more negative description of a person. The methylation study examined this subgroup because treatment history and clinical course could influence molecular measurements.

Many schizophrenia-associated methylation differences were especially evident in the treatment-resistant group. Clozapine exposure was one possible contributor considered by the authors. The measured pattern therefore could contain treatment-related effects as well as disease-associated or exposure-associated effects.

A treatment can change cellular regulation while reducing symptoms. Detecting its molecular footprint does not make that footprint the original cause of illness. Clinical subgroup comparisons need to preserve the sequence of disease, medication exposure, and sample collection.



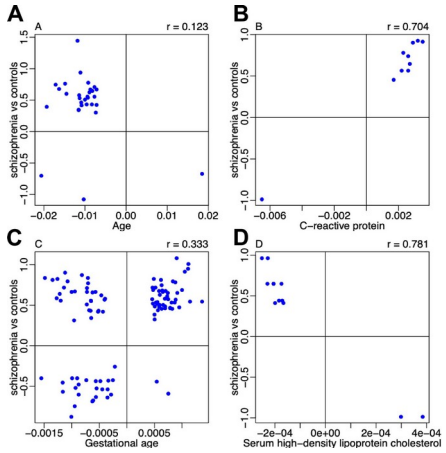
Hannon et al. (2021), Fig. 8.

Replication across cohorts tests the stability of a signature

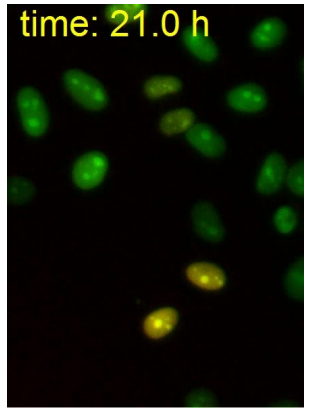
Different cohorts can vary in recruitment, laboratory processing, medication exposure, and environmental history. Combining them requires careful handling of those differences. The study used a common analytical framework to compare methylation results across independent datasets.

Some methylation sites showed evidence of overlap with regions identified by genetic-association studies. This placed sequence variation and chemical regulation near some of the same genomic regions. It did not mean that every genetic association acted through the same methylation change.

Cellular state can persist across division. In a separate fluorescent reporter preparation, most daughter cells inherit the parent's high or low expression state, although some daughters diverge. The recording follows expression within cell lineages, whereas the clinical blood analysis compares sampled molecular signatures across people.



Hannon et al. (2021), Fig. 6.



Expression states inherited across cell divisions. Tan et al. (2021).

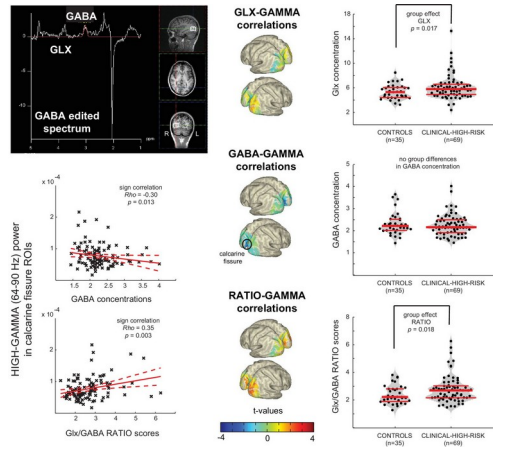
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Excitation and inhibition organize cortical timing

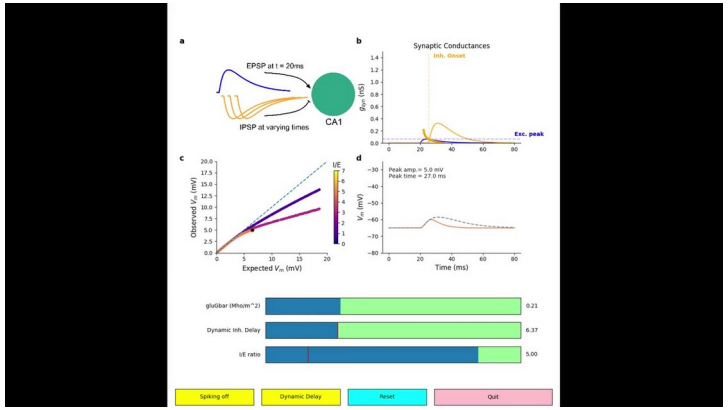
Excitatory neurons recruit other cells, while inhibitory interneurons constrain when and how strongly those cells respond. A functioning cortical network depends on the timing and distribution of both actions. Balance does not mean equal firing rates in every cell population.

Gamma-band activity is a fast rhythm produced through interactions among neural populations. Its amplitude can change when excitation or inhibition changes, but it is not a direct count of inhibitory synapses. Several cellular arrangements can produce a similar recorded rhythm.

A single-cell computer model varies the delay between excitation and inhibition. When stronger excitation brings earlier inhibition, the combined voltage response grows less steeply. Holding that delay fixed changes the response curve, showing how timing can alter a neuron's input-output behavior without changing its identity.



Grent-'t-Jong et al. (2018), Fig. 5.



Model: excitation and inhibition timing. Bhatia et al. (2019).

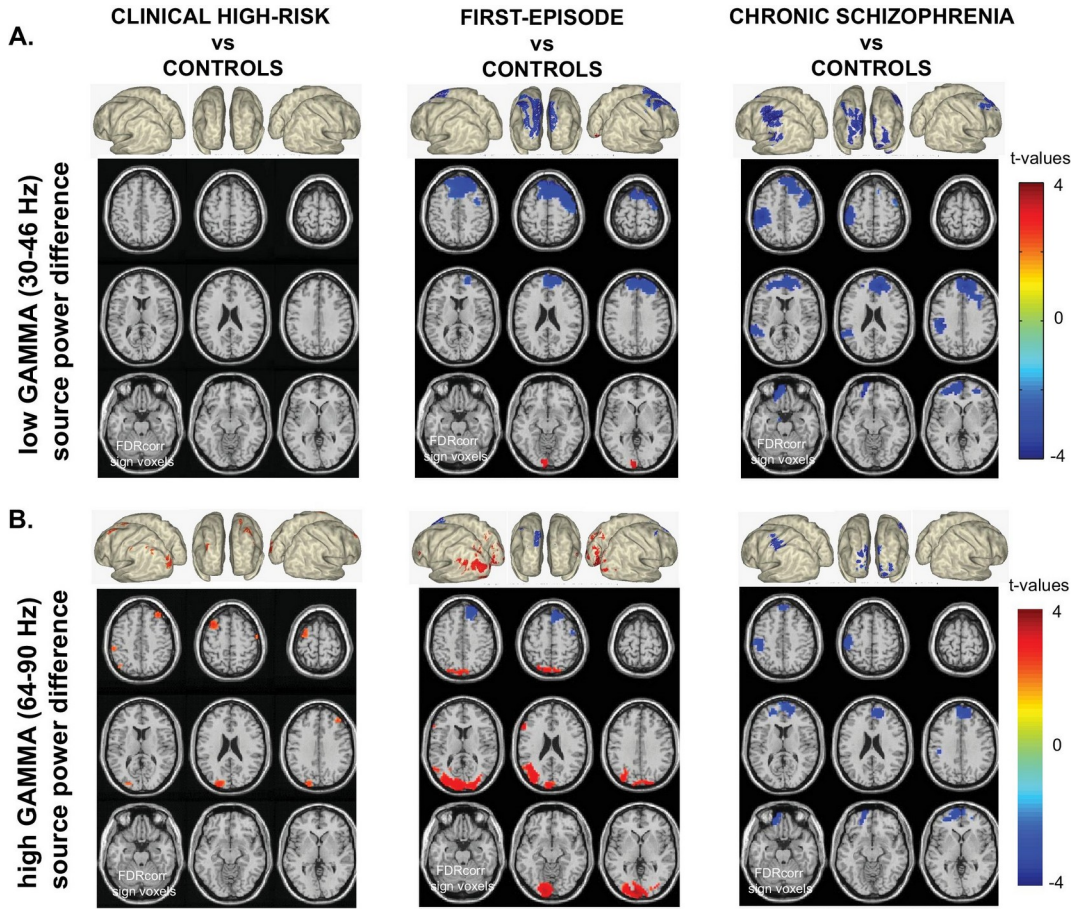
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Gamma activity changes across clinical stages

Researchers measured resting MEG activity in people at clinical high risk, people experiencing a first episode, people with chronic schizophrenia, and comparison participants. The groups represented different clinical stages rather than a single homogeneous disease sample.

High-risk participants showed increased high-frequency gamma power, while first-episode and chronic groups had different patterns involving lower and higher gamma frequencies. The direction of a group difference therefore depended on clinical stage and the frequency range examined.

This was a comparison of groups, not a longitudinal record of every person moving through those stages. The results describe stage-associated patterns and do not establish that every high-risk person's activity will evolve into the pattern observed in the chronic group.



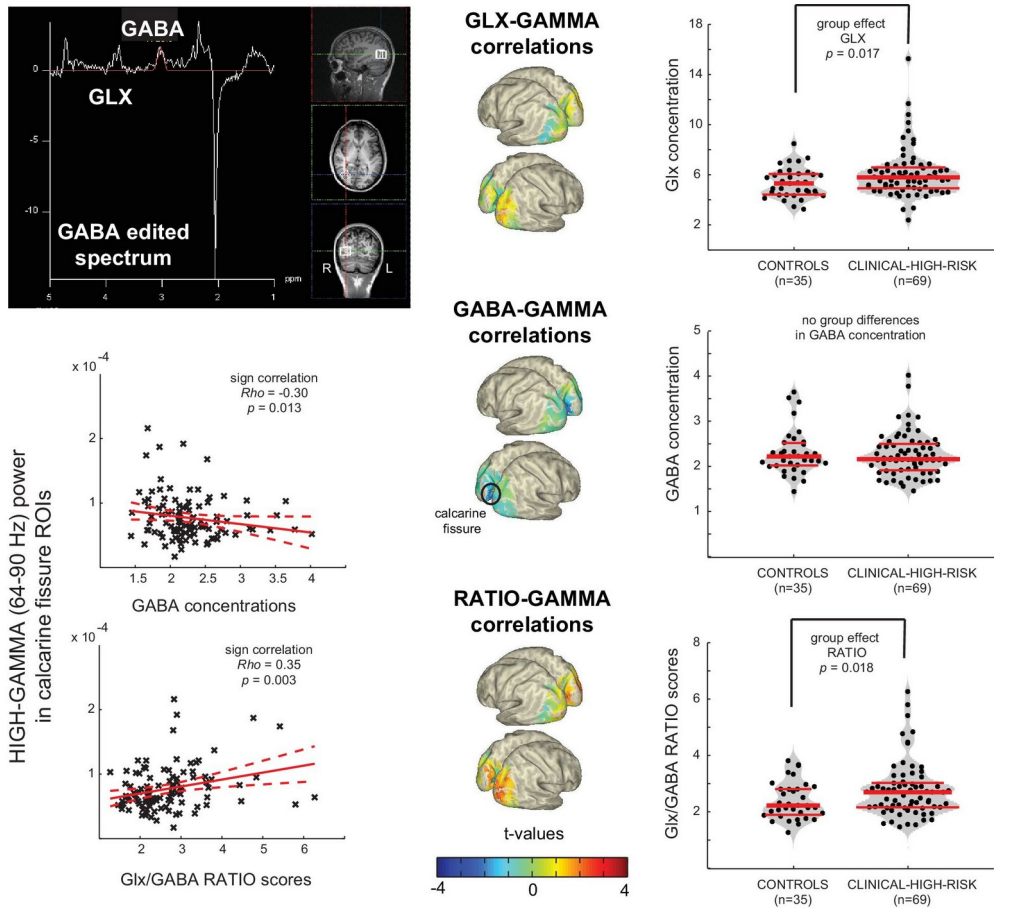
Grent-'t-Jong et al. (2018), Fig. 1.

Spectroscopy measures tissue metabolites

Magnetic resonance spectroscopy estimates concentrations of selected chemicals within a tissue volume. In the high-risk group, the investigators compared glutamate-related and GABA-related measurements with gamma activity. These metabolites participate in excitation and inhibition, respectively.

A tissue concentration is not the same as transmitter release at a particular synapse. The measurement includes chemical pools across many cells and compartments. Its interpretation therefore remains different from measuring the response of one postsynaptic neuron.

The study found an association between the metabolite balance and occipital gamma power in the high-risk group. The two measurements sampled related network biology at different scales, providing complementary observations without making either a direct measurement of every excitatory and inhibitory current.



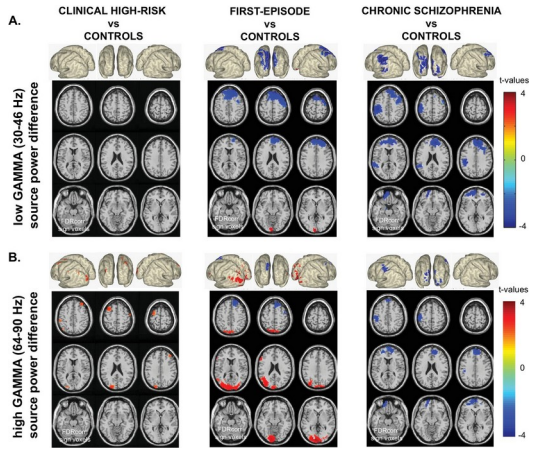
Grent-'t-Jong et al. (2018), Fig. 5.

Cognitive symptoms require precise task measurements

Working memory maintains information briefly so it can guide an ongoing action. Attention selects relevant information, while cognitive control helps maintain a goal despite distraction. These functions involve overlapping networks but can be impaired to different degrees.

A task score combines several processes, including perception, motivation, movement, and understanding of instructions. A low score therefore needs interpretation within the task's demands. It cannot automatically be assigned to failure of one receptor or one brain region.

A moving-dot task can separate motion direction from the reliability of sensory evidence. The demonstration shows low- and high-volatility stimulus sequences. Its introductory labels explain stimulus conditions to the viewer; the participants judged the moving dots without seeing these added explanatory labels.



Grent-'t-Jong et al. (2018), Fig. 1.



Moving-dot task stimuli. Zylberberg et al. (2016).

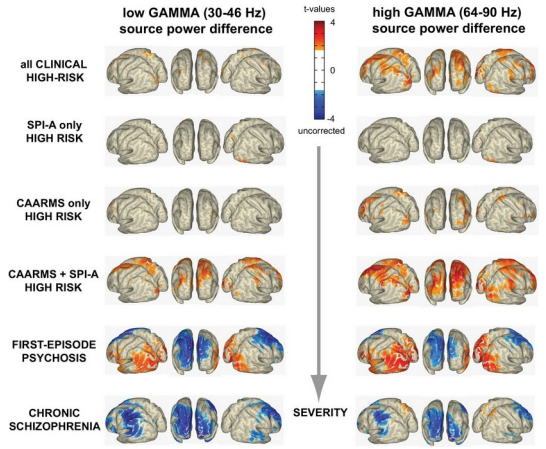
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The same rhythm can relate differently to symptoms

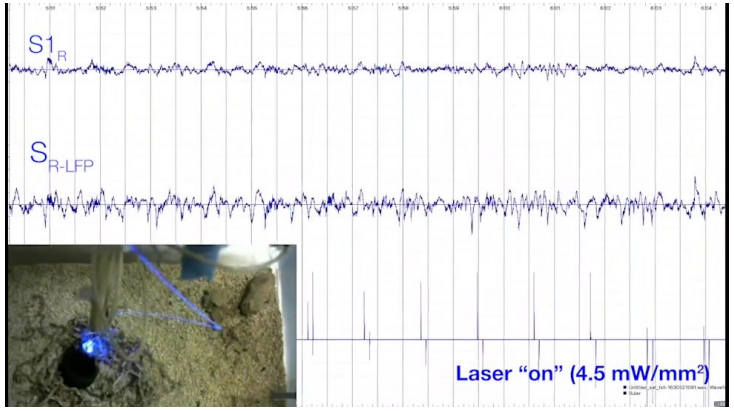
In first-episode and chronic schizophrenia groups, reduced gamma activity was associated with aspects of cognitive impairment and clinical symptoms. In the high-risk group, elevated psychotic symptoms showed a different association with gamma power. The clinical context changed the observed pattern.

A single rule such as more gamma means more illness would therefore misdescribe the results. Frequency, region, and clinical stage all identified specific measured conditions. The study analyzed those conditions rather than averaging them into one undifferentiated abnormality.

Inhibition can also be tested causally in a different preparation. Activating light-sensitive parvalbumin interneurons suppresses brief abnormal electrical discharges in a myelin-deficient mouse. The recording follows that intervention and its voltage signals, rather than the clinical study's relationship between gamma activity and symptom severity.



Grent-'t-Jong et al. (2018), Fig. 3.



Interneuron activation suppressing abnormal discharges. Dubey et al. (2022).

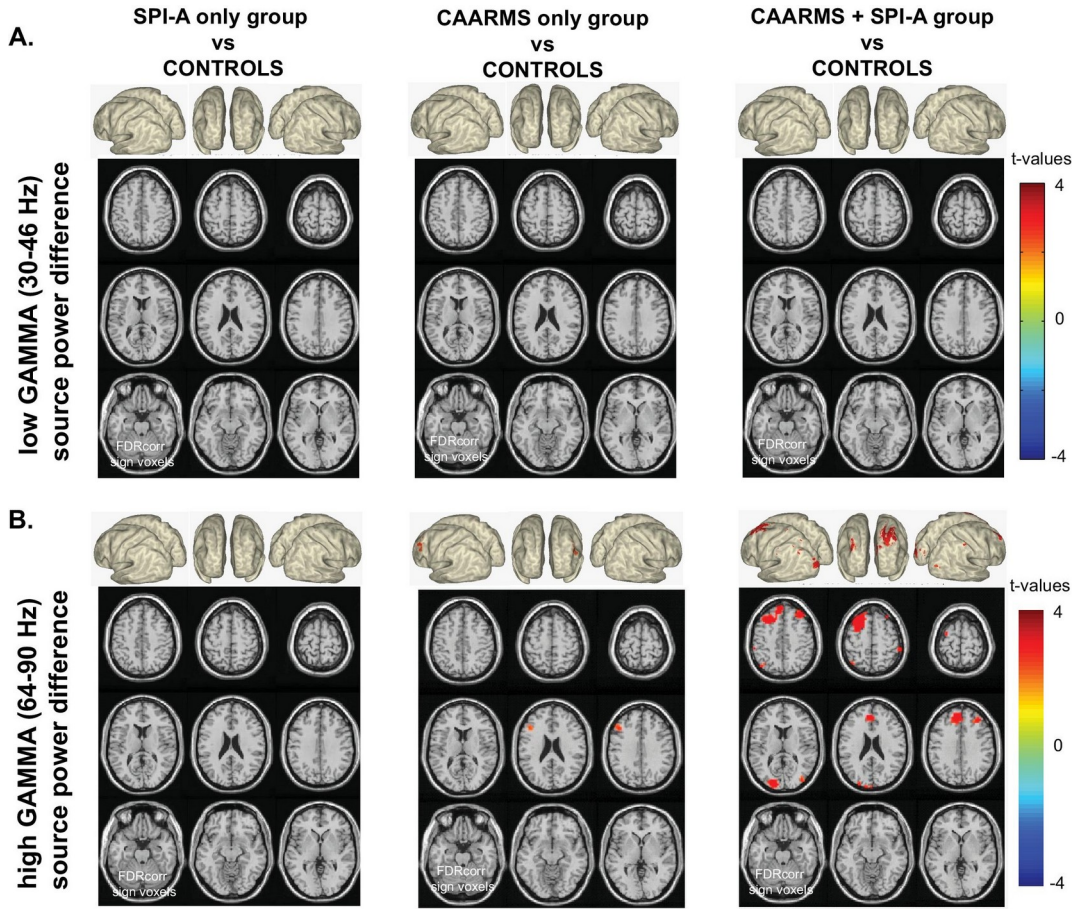
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Resting measurements capture a state of the network

Resting-state activity is measured without a specific experimental task requiring repeated responses. It still depends on arousal, ongoing thought, sensory conditions, and the person's physiological state. Resting does not mean that the brain is inactive.

The MEG study estimated source-level spectral power to compare activity across regions and frequencies. Source estimates use a model to infer where recorded fields originated. Their spatial scale differs from that of a microelectrode placed beside a single neuron.

The observed gamma patterns therefore describe a measured population state under defined recording conditions. They can be compared across groups and with symptoms while remaining distinct from a microscopic explanation of which cells generated the altered activity.



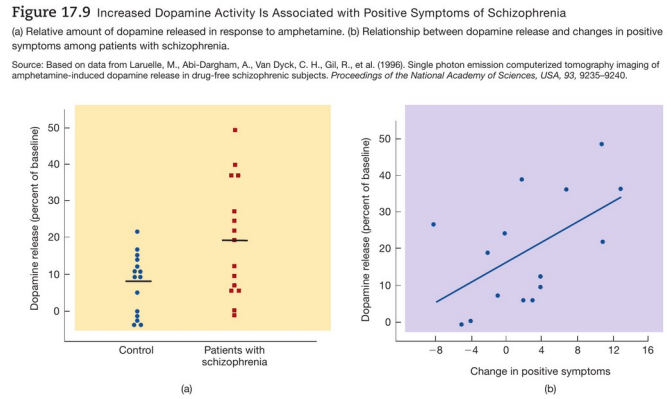
Grent-'t-Jong et al. (2018), Fig. 2.

Dopamine changes the significance of signals and actions

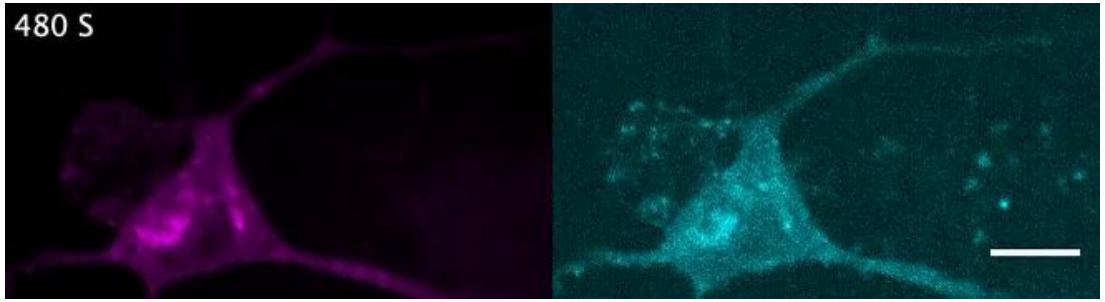
Dopamine influences learning, motivation, and selection of behavior through several pathways. The mesolimbic pathway projects from the ventral tegmental area toward regions including the ventral striatum. It is involved in assigning significance and updating expectations about outcomes.

Excessive or mistimed dopamine signaling can contribute to psychotic symptoms, but schizophrenia is not explained by uniformly high dopamine everywhere. Cortical, striatal, and other circuits have different cellular properties, inputs, and patterns of activity.

Dopamine can activate different receptor classes on different cells. Separate microscopy follows a receptor-state sensor in D1-receptor-expressing striatal neurons exposed to dopamine. This cellular preparation examines D1 signaling, while the antipsychotic mechanism discussed here chiefly concerns reducing signaling through D2 receptors.



Carlson and Birkett (2022), Fig. 17.9, p.560.



D1 receptor-state sensor in striatal neurons. Puri et al. (2022).

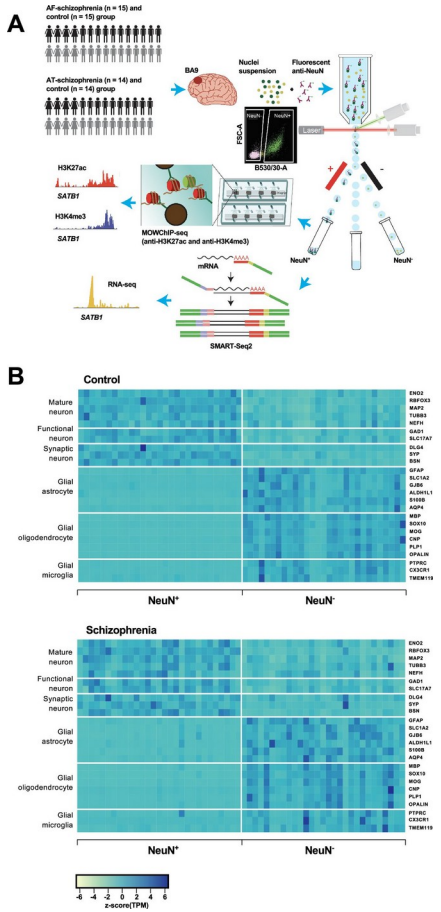
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Medication can change chromatin in frontal cortex

Chromatin consists of DNA associated with proteins called histones. Chemical modifications of histones can change access to gene-regulatory regions. Researchers examined such modifications and RNA expression in postmortem frontal cortex from people with schizophrenia and matched comparison individuals.

They separated neuronal and non-neuronal samples and distinguished people with different antipsychotic treatment histories. This design addressed whether a molecular difference might accompany illness, treatment, or both. The cellular origin of the material was part of the analysis.

The study found extensive epigenomic differences, including some altered features associated with treatment exposure. A postmortem comparison records a person's accumulated biological history, rather than experimentally assigning treatment while holding every other lifetime influence constant.



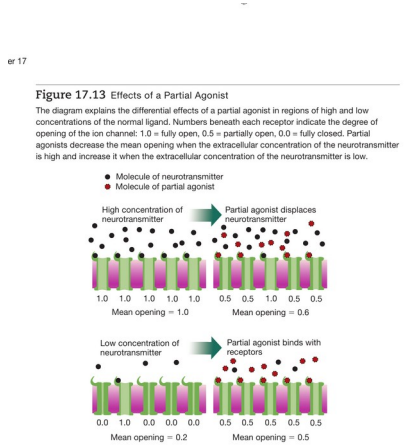
Zhu et al. (2024), Fig. 1.

D2 blockade and partial agonism act differently

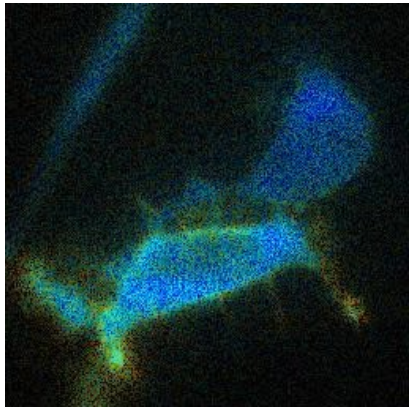
An antagonist occupies a receptor and reduces activation by the normal transmitter. A partial agonist can activate the receptor but produces a smaller response than a full agonist under comparable conditions. Its effect therefore depends partly on the surrounding transmitter environment.

Where dopamine activity is high, a partial agonist can compete with dopamine and reduce the average receptor response. Where dopamine activity is low, the same drug can supply some receptor activation. This is a pharmacological property, not a promise of identical effects across every brain region.

A separate cell-based experiment measures signaling after dopamine activates D2 receptors. A light-emitting molecular sensor changes its signal as a signaling partner approaches the membrane. This preparation measures dopamine-driven receptor signaling; it does not directly compare an antagonist with the partial agonist aripiprazole.



Carlson and Birkett (2022), Fig. 17.13, p.564.



D2 activation and membrane signaling sensor. Avet et al. (2022).

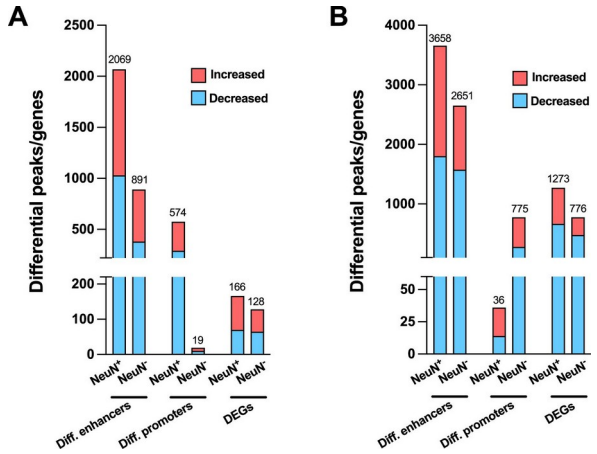
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Illness and medication leave different marks

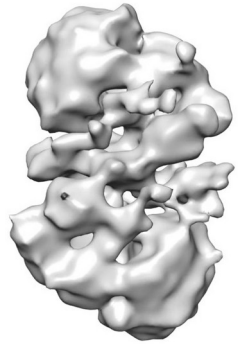
The frontal-cortex study compared antipsychotic-free and antipsychotic-treated groups with matched controls. Some regulatory differences appeared to be reversed in treated samples, while others were associated with medication exposure. A treatment footprint could therefore differ from the initial illness-associated pattern.

The investigators examined marks associated with active promoters and enhancers. Promoters help initiate transcription near a gene, while enhancers regulate expression through interactions that can act over longer genomic distances. These marks described regulatory states rather than DNA-sequence changes.

Chromatin regulation also involves enzymes that remove acetyl groups from histones, the proteins around which DNA is wrapped. A separate electron-microscopy structure shows HDAC1 within a larger regulatory complex. The rotating structure locates protein components rather than showing medication changing gene expression in a living brain.



Zhu et al. (2024), Fig. 6.



HDAC1-containing chromatin-regulatory complex. Millard et al. (2016).

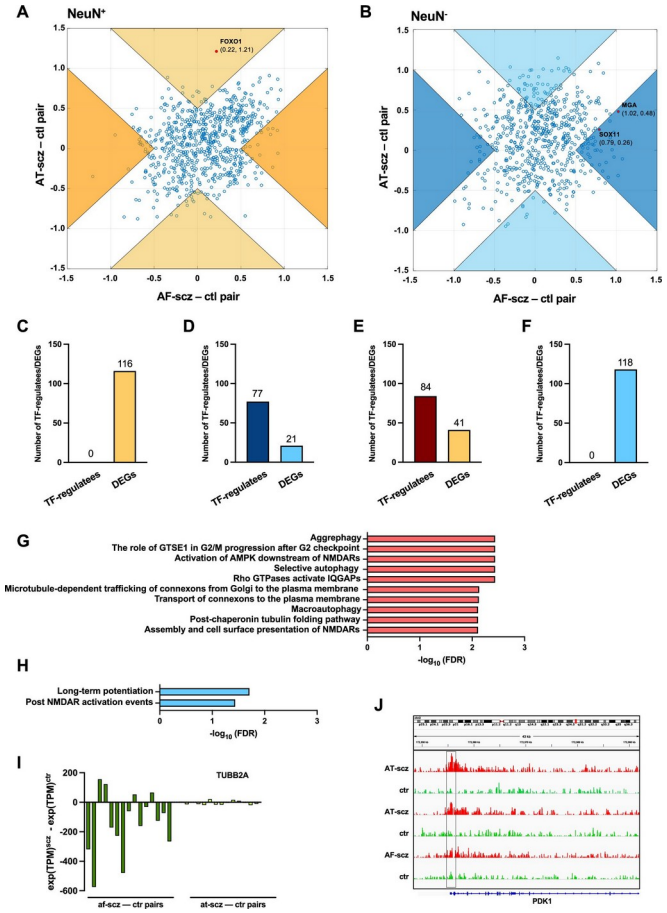
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Treatment benefits require separate measurements

Reducing hallucinations, preventing relapse, improving daily functioning, and reducing distress are related but distinct goals. A medicine can improve one outcome while causing adverse effects in another domain. Clinical follow-up therefore measures both symptom change and the person's experience of treatment.

Dopamine-receptor actions outside the desired pathway can affect movement, hormone regulation, and other functions. Drugs also differ in their actions at additional receptors. These pharmacological differences help explain why tolerability and benefit vary among treatments and among people.

Medication is one component of care alongside psychological and social interventions. Access to housing, relationships, education, and practical support can change functioning without being reducible to receptor occupancy. The clinical outcome concerns the person's life as well as a symptom score.



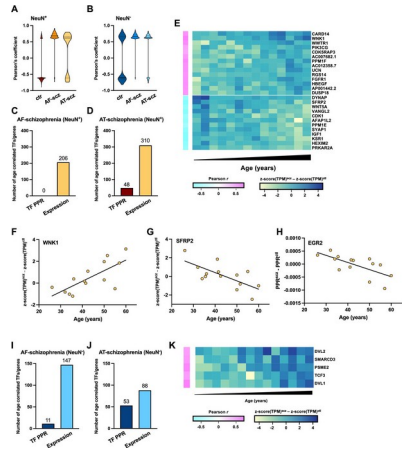
Zhu et al. (2024), Fig. 5.

Cell-specific measures reveal different regulatory responses

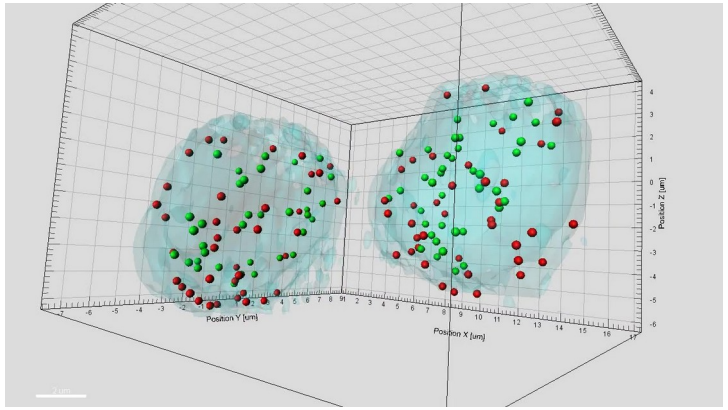
Neuronal and non-neuronal nuclei contain the same overall genome but use different regulatory programs. Separating these populations allowed the investigators to identify changes that would be obscured in a mixed-tissue average. A neuron-associated result was not assigned automatically to glia.

The study also examined age in relation to epigenomic patterns. Treatment history and age were not interchangeable variables, and their effects could overlap. This matters when postmortem groups differ in both lifetime exposure and disease duration. Chromatin accessibility describes how easily regulatory proteins can reach a DNA region within its surrounding protein packaging.

Nuclear organization changes during cell division. Separate microscopy follows a dividing mouse embryonic stem cell, with labels identifying the nuclear envelope and different chromatin regions. The depth reconstruction depicts physical chromosome organization, a different process from the cortical cell-type-specific gene regulation measured in the clinical samples.



Zhu et al. (2024), Fig. 7.



Chromatin organization during cell division. Poleshko et al. (2019).

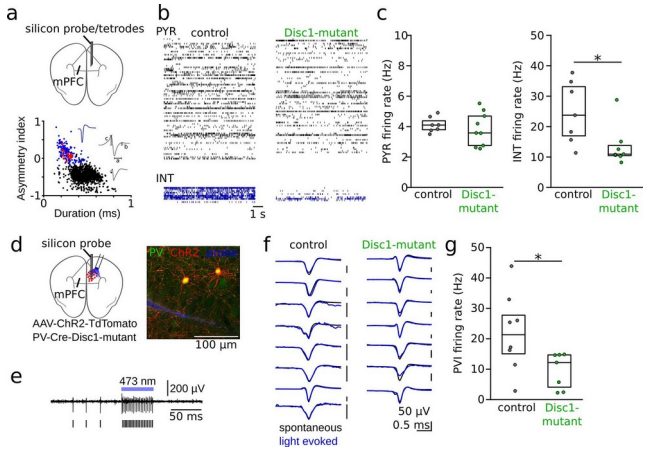
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Prefrontal circuits depend on recruited inhibition

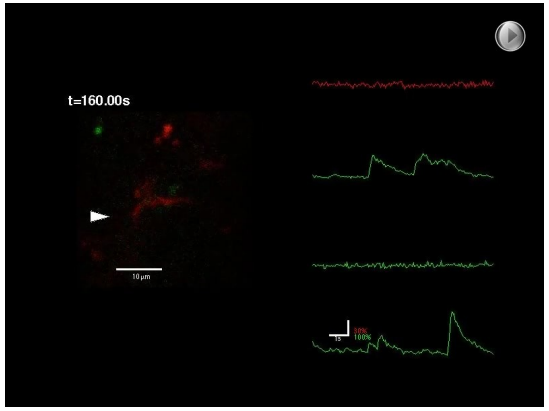
Pyramidal neurons provide much of the cortex's excitatory output. They also recruit local interneurons that feed inhibition back into the network. This arrangement regulates the timing and strength of activity while a cortical population represents information.

Parvalbumin interneurons can fire rapidly and influence many nearby pyramidal cells. Their activity depends on receiving sufficient excitatory drive. A reduction in interneuron firing can therefore arise from weakened input, altered intrinsic properties, or both.

Inputs and receiving branches can be imaged separately using two calcium-sensitive colors. A mouse visual-cortex recording labels incoming axons and pyramidal-cell dendrites in the same superficial layer. Their signals reveal activity in distinct cellular compartments rather than counting every fluorescent change as firing by one cell class.



Sauer and Bartos (2022), Fig. 1.



Two-color axonal and dendritic calcium imaging. Dana et al. (2016).

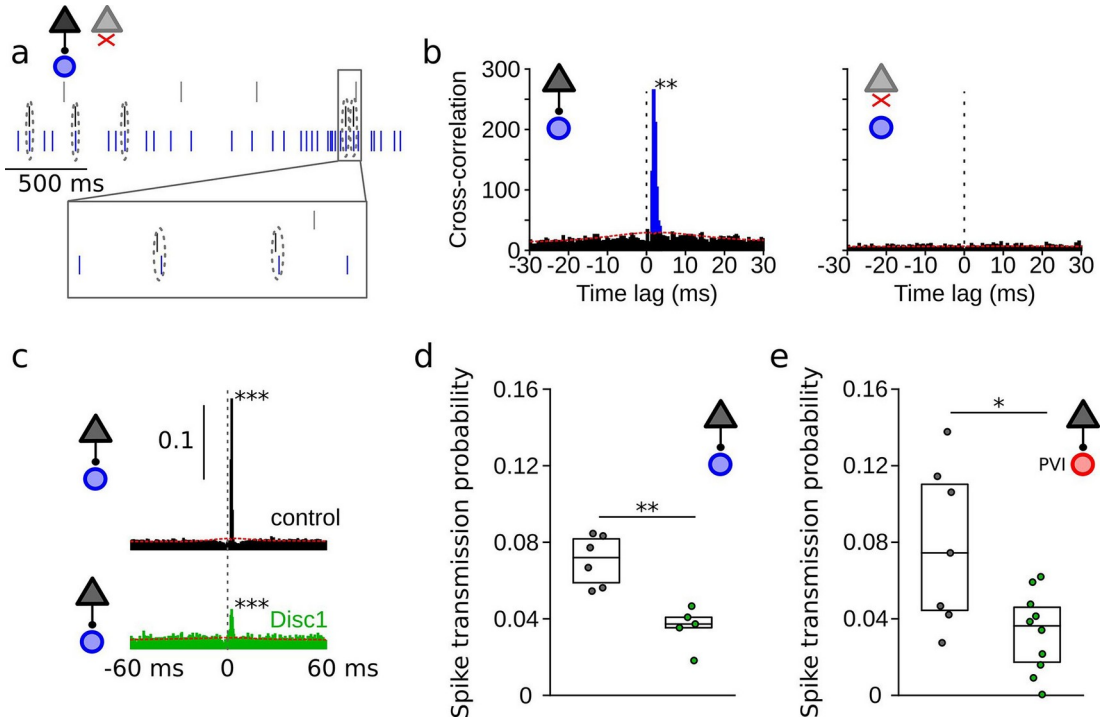
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Disc1 mutation weakens local recruitment of interneurons

Researchers recorded prefrontal neurons in awake mice carrying a Disc1 mutation. DISC1 is a gene historically studied in psychiatric-risk biology. The mouse manipulation provides a controlled way to examine its circuit functions, rather than reproducing the full human diagnosis.

Fast-spiking interneurons, including identified parvalbumin cells, had reduced average firing in the mutant mice. The investigators also measured communication between local pyramidal neurons and interneurons. This tested the synaptic recruitment step rather than assuming that reduced firing meant interneurons were absent.

Spike transmission from pyramidal cells to interneurons was less effective. The result supplied a route through which altered excitatory input could reduce inhibitory-cell participation, changing how the local prefrontal network organizes its activity.



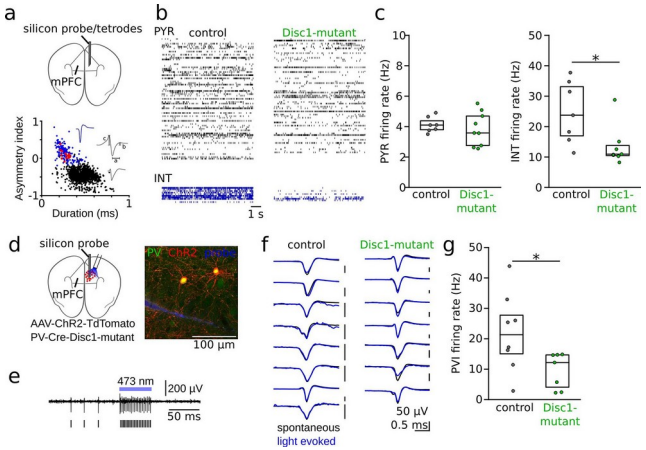
Sauer and Bartos (2022), Fig. 2.

Optogenetic identification ties spikes to a cell class

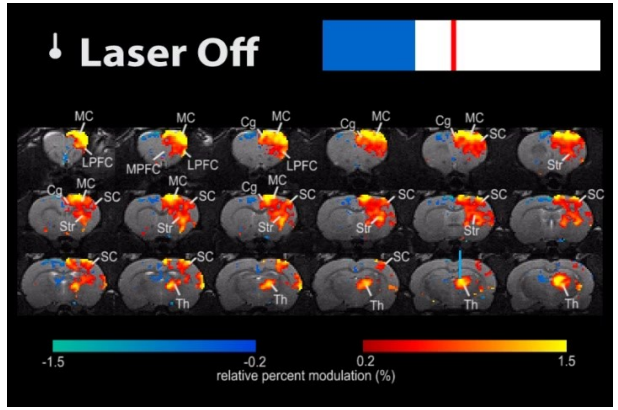
A recorded spike does not always reveal the cell's molecular identity. The investigators used optogenetic identification for parvalbumin neurons: light-sensitive proteins allowed a defined cell class to respond to light, helping connect recorded activity with that class.

This identification strengthened the interpretation of reduced firing in fast-spiking cells. Fast spike shape alone is a physiological clue, while a cell-class manipulation provides additional information. The two forms of identification answered related but different questions.

Optogenetics can also be combined with brain-wide imaging. A separate experiment stimulates central-thalamic excitatory neurons at forty pulses per second and follows changing MRI signals across regions. These delayed blood-based signals measure a different consequence from the individual spikes used to identify interneurons.



Sauer and Bartos (2022), Fig. 1.



Thalamic stimulation and brain-wide MRI signals. Liu et al. (2015).

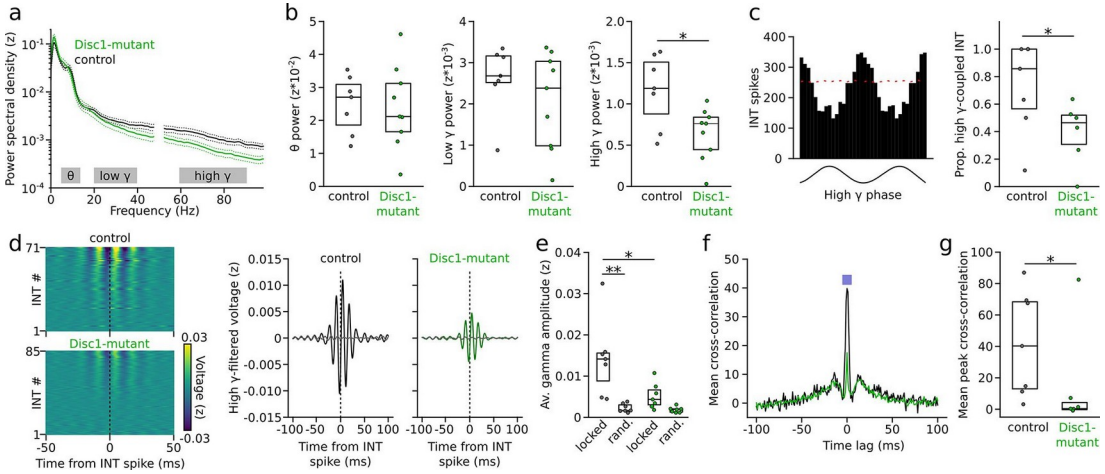
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Cell assemblies represent coordinated population activity

A cell assembly is a group of neurons whose activity tends to occur together within a relevant time window. Assemblies can participate in representing an event or guiding an action. Their organization depends on both excitatory recruitment and inhibitory control.

The same average firing rate can coexist with different patterns of coactivation. Neurons may fire the same total number of spikes while grouping those spikes differently in time. A population analysis therefore supplies information that a mean rate alone does not capture.

Gamma-phase coupling is another timing measure. It describes whether a cell tends to fire at particular phases of a population rhythm. Assembly structure and phase coupling examine different forms of temporal organization within the same network.



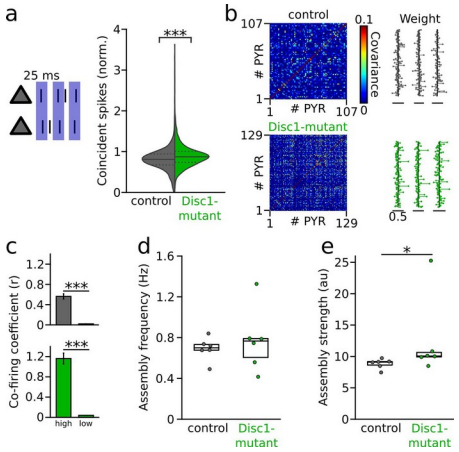
Sauer and Bartos (2022), Fig. 4.

Reduced inhibition changes assembly strength

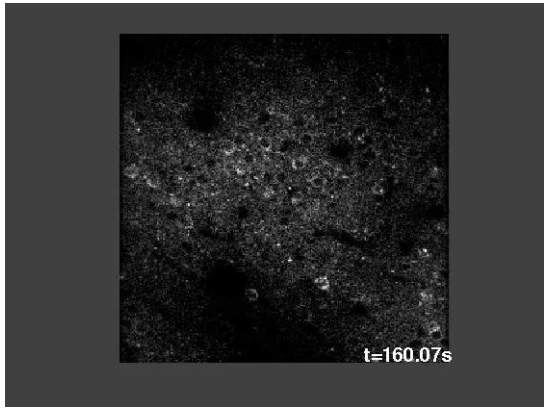
The Disc1 study examined how groups of pyramidal neurons activated together. Assembly activations were not simply absent in mutant mice. Instead, the expression strength of individual patterns increased, even though the overall occurrence of assemblies was not increased in the same way.

Interneurons also showed reduced phase coupling to ongoing gamma activity. This placed a cell-type-specific timing abnormality alongside altered pyramidal-cell organization. The measurements described how activity was coordinated rather than only whether the cortex was globally active.

A separate visual-cortex recording follows individual neurons responding to moving gratings. Calcium-sensitive fluorescence changes as stimulus directions alternate with blank periods. The preparation measures sensory responses in anesthetized mice rather than the mutant prefrontal circuit's inhibitory connections or its memory-related assemblies.



Sauer and Bartos (2022), Fig. 5.



Visual-cortex responses to moving gratings. Dana et al. (2016).

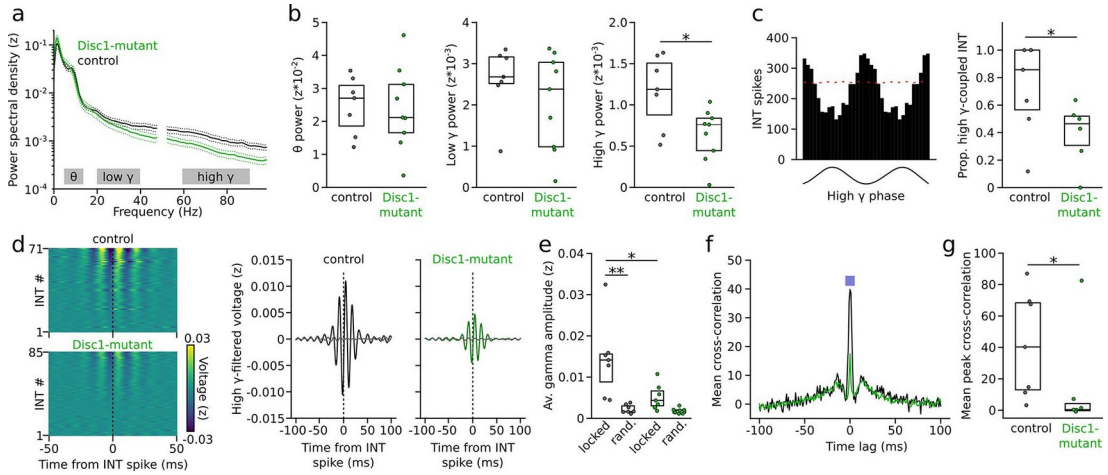
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A gene model isolates a circuit function

Changing Disc1 in mice allows investigators to manipulate a defined biological factor. The measured outcome was prefrontal communication and population organization. Human schizophrenia involves many risk variants, exposures, and clinical features that are not all represented by this manipulation.

The experiment nevertheless resolves a concrete interaction: local glutamatergic input recruited interneurons less effectively. That observation can guide further tests of which synaptic properties account for the reduction and how altered recruitment affects a particular cognitive task.

The explanatory scale remains explicit. A synaptic communication defect in awake mouse cortex is evidence about circuit function. It becomes a claim about a human symptom only when additional work measures the corresponding circuit and behavior in an appropriate clinical context.



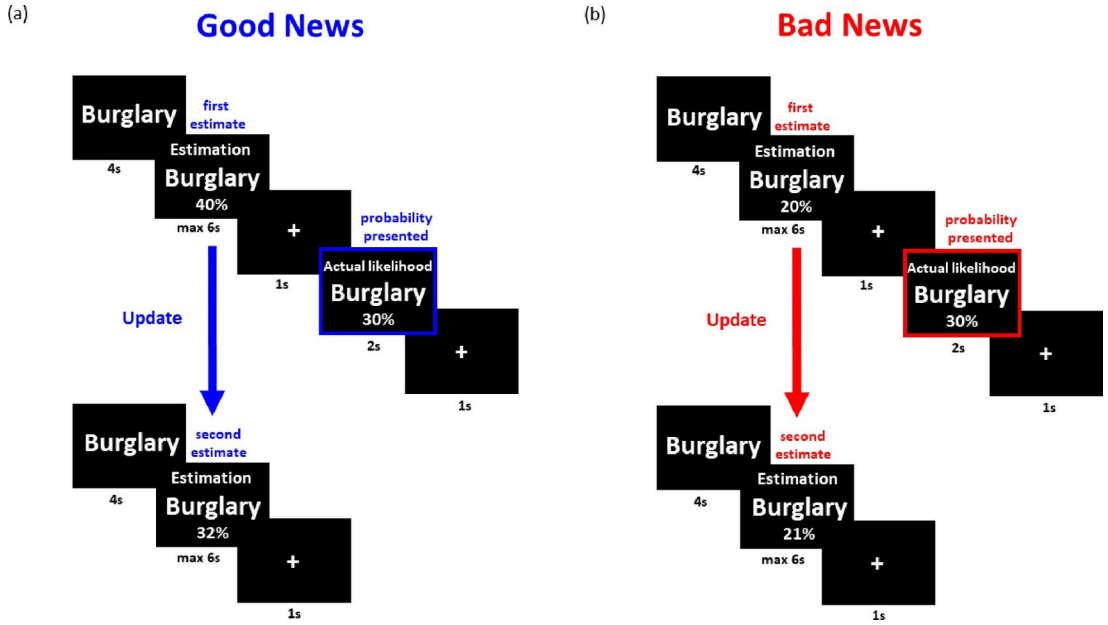
Sauer and Bartos (2022), Fig. 4.

Mood disorders change experience and biological rhythms

Major depression can involve persistent low mood, reduced pleasure, altered sleep and appetite, changes in movement, and impaired concentration. A clinical episode is more than an ordinary fluctuation in feeling. Its duration, severity, context, and effects on functioning are assessed together.

Bipolar disorders include episodes of elevated or irritable mood with increased energy and activity, alongside patterns that can include depression. Mania can involve decreased need for sleep, rapid speech, impulsivity, and impaired judgment. Hypomania differs in severity and consequences.

Psychotic symptoms can occur during mood episodes, so psychosis alone does not distinguish schizophrenia from a mood disorder. The timing of psychosis relative to mood changes and the broader clinical course help establish the diagnostic pattern.



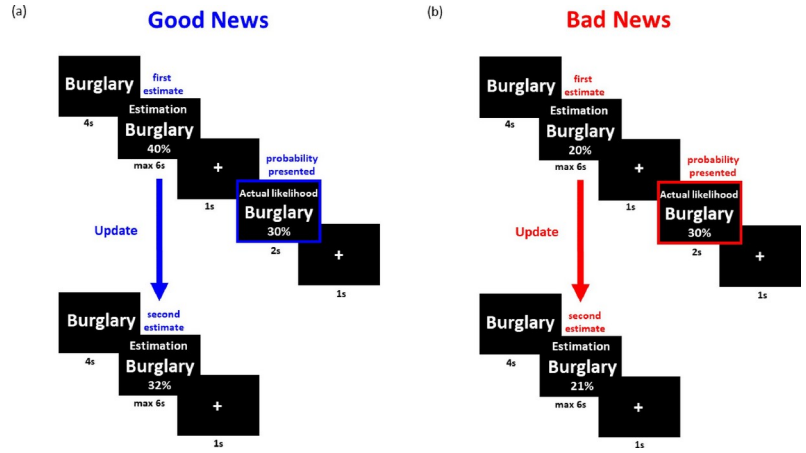
Ossola et al. (2020), Fig. 1.

Belief updating can be measured between bipolar episodes

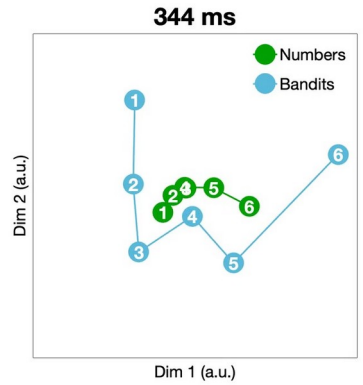
Euthymia is a period without a current depressive or manic episode. Researchers studied belief updating in people with bipolar disorder during euthymia, then followed their clinical course. This design examined whether a measured learning pattern preceded later recurrence.

Participants received positive and negative information and had opportunities to revise beliefs. The task quantified how much each kind of information changed an estimate. The measurement therefore separated responses to favorable and unfavorable information rather than assigning one overall learning score.

Learning tasks can also be compared through patterns of brain activity. A separate human imaging study projects changing activity from two probability tasks into a shared coordinate space. The trajectories represent relationships among measured neural patterns, rather than the actual screen display or a person's bipolar symptoms.



Ossola et al. (2020), Fig. 1.



Neural-pattern trajectories during two human tasks. Luyckx et al. (2019).

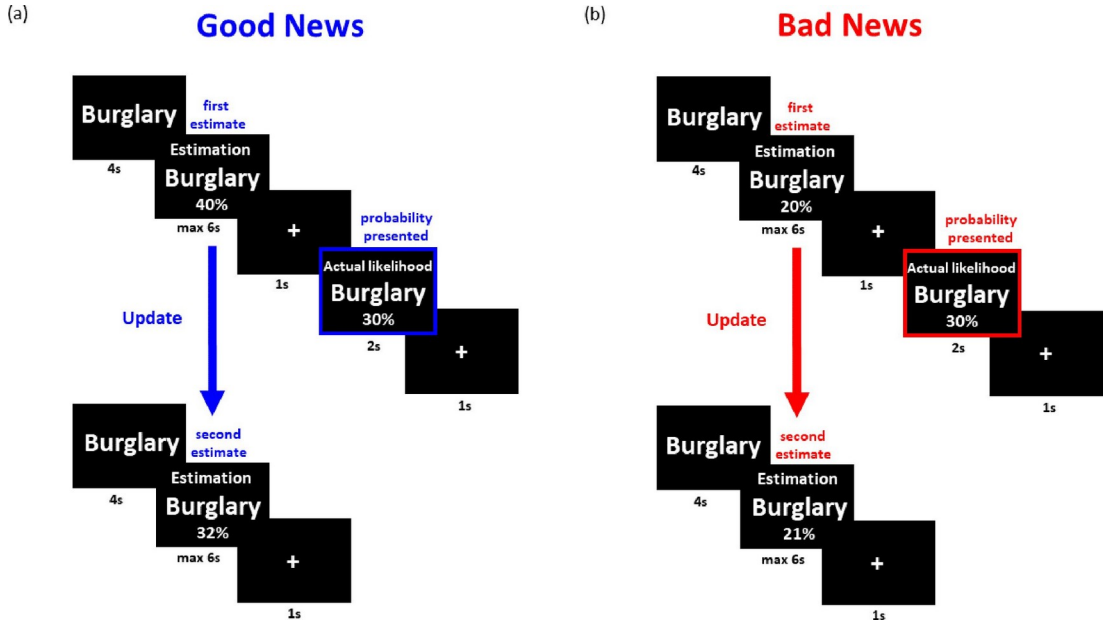
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Mania changes sleep need and goal-directed behavior

During mania, a person may remain highly active while sleeping much less than usual. Decreased need for sleep differs from wanting to sleep but being unable to do so. This distinction helps characterize the biological and behavioral state of an episode.

Increased goal-directed activity can become difficult to regulate. Plans may expand rapidly, attention may shift, and decisions can produce serious consequences. The clinical concern is the change from the person's usual functioning and the resulting impairment or danger.

Depression and mania are not simply opposite positions on one scale of happiness. They involve changes in energy, sleep, cognition, motivation, and regulation. A person can also experience combinations of symptoms that do not fit an uncomplicated happy-versus-sad description.



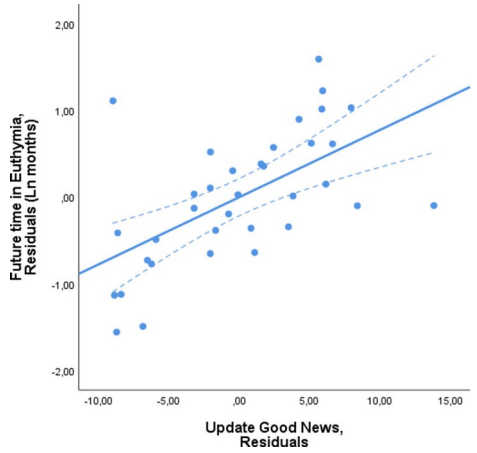
Ossola et al. (2020), Fig. 1.

Learning separates favorable and unfavorable evidence

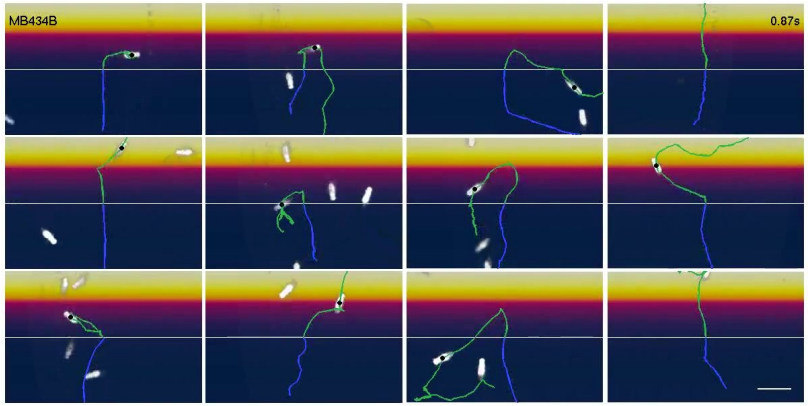
Valence describes whether information or an outcome is favorable or unfavorable. In the bipolar study, belief revision was analyzed separately for positive and negative information. This revealed an imbalance that would be obscured by averaging all updates together.

Reduced use of favorable information can sustain more pessimistic expectations during an otherwise well period. The study observed that this pattern predicted earlier recurrence. The proposed psychological route concerns how expectations evolve as new events are encountered.

Valence can be manipulated in simpler circuits. In a separate fly experiment, activating selected mushroom-body output neurons makes animals avoid illuminated areas. Tracked paths show several ways of turning away, so the measured preference does not require one fixed movement or a human-like report of mood.



Ossola et al. (2020), Fig. 3.



Fly choices during output-neuron activation. Aso et al. (2014).

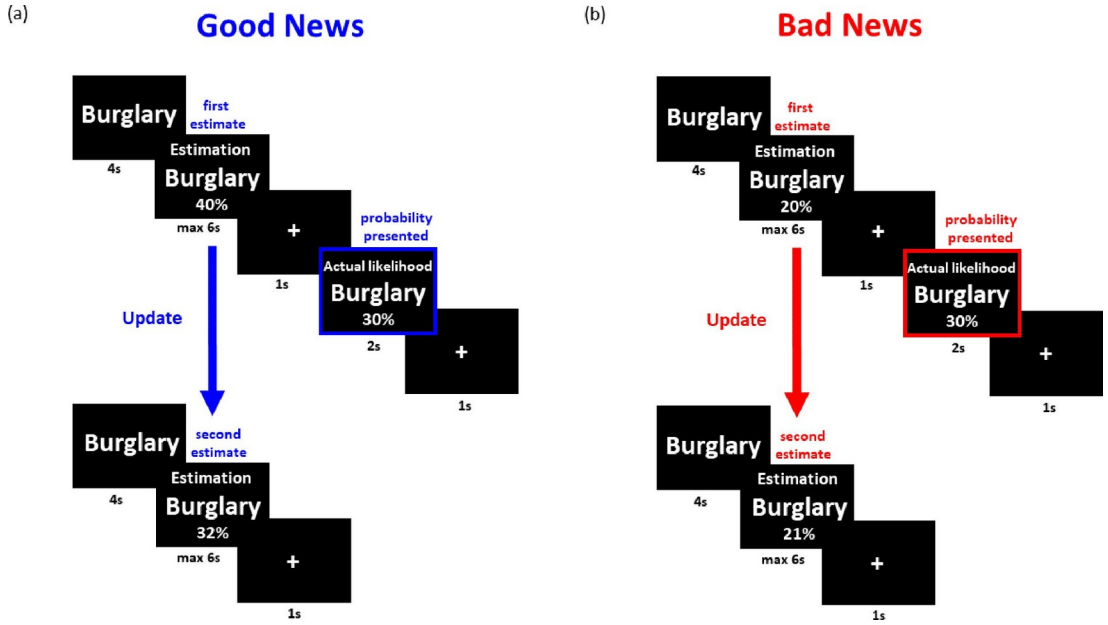
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Inherited vulnerability does not specify an episode's timing

Bipolar disorder has a genetic contribution distributed across multiple biological pathways. Inherited vulnerability persists during periods of wellness, but episodes occur at particular times. Risk factors and immediate triggers therefore occupy different positions in the disease course.

Sleep disruption, changes in routine, medical conditions, and treatment interruption can be relevant to relapse, depending on the person. Their presence does not make every episode predictable. Monitoring focuses on patterns that have actually preceded changes for that individual.

Long-term management aims to reduce recurrence and preserve functioning between episodes. It evaluates mood, sleep, medication effects, and the person's daily circumstances. A laboratory learning task supplies one additional observation rather than replacing this broader clinical assessment.



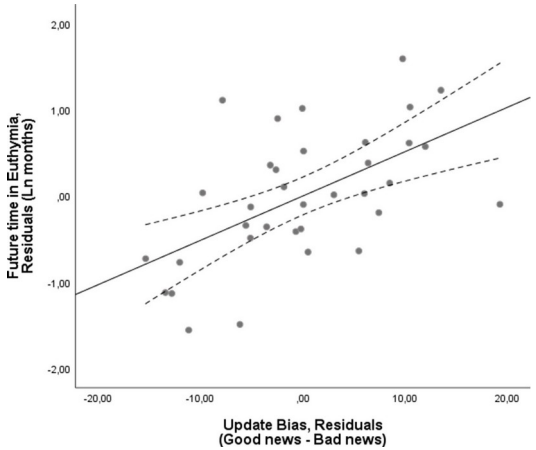
Ossola et al. (2020), Fig. 1.

A predictor needs a separate causal test

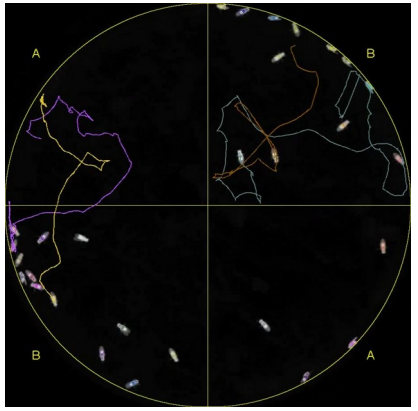
A marker predicts an outcome when it provides information about what happens later. It becomes a demonstrated treatment mechanism only when changing it alters the relevant outcome under a suitable test. The belief-updating study addressed the first question.

The favorable-versus-unfavorable learning imbalance was measured before recurrence. The investigators could therefore relate it to time until another episode. They did not randomly assign participants to have one learning pattern or another during the follow-up period.

A causal learning experiment can pair an odor with activation of a defined fly neuron. During the later test, flies avoid the trained odor while crossing toward the comparison odor. This intervention changes a specific learned choice rather than establishing a treatment for relapse in bipolar disorder.



Ossola et al. (2020), Fig. 2.



Fly odor avoidance after optogenetic training. Aso et al. (2016).

Play video

Monoamine medicines act quickly at transporters

Serotonin and norepinephrine are monoamine transmitters. After release, transporters help remove them from extracellular space. Many antidepressants inhibit these transporters, increasing the time transmitter remains available to act on receptors near the synapse.

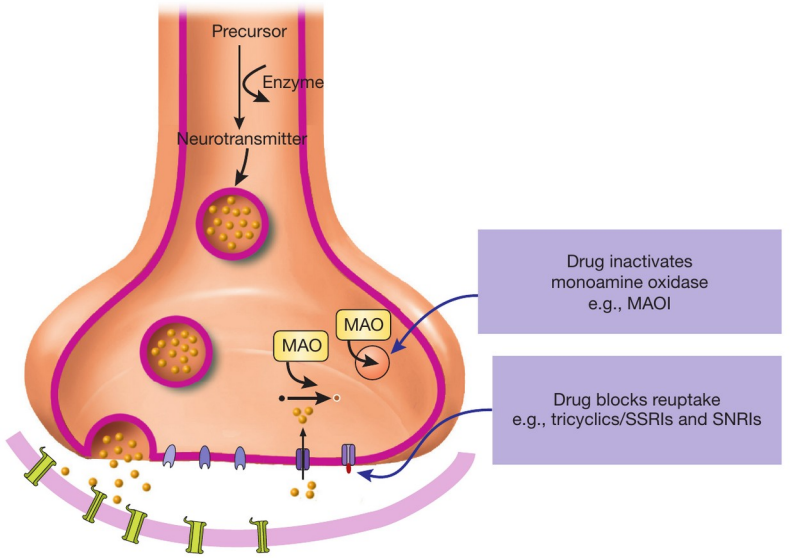
This pharmacological action begins before the full clinical response usually appears. The delay means that transporter occupancy alone is not a complete explanation of symptom improvement. Receptor adaptation, gene regulation, synaptic change, and network activity can develop over different timescales.

Monoamine oxidase inhibitors act through another route by reducing enzymatic breakdown of monoamines. Different drug classes therefore change transmitter availability through different cellular steps. Their eventual clinical effects still need direct measurement rather than inference from transmitter concentration alone.

er 17

Figure 17.14 Pharmacological Treatment Strategies for Depression

Treatment strategies include decreasing enzymatic deactivation or reuptake of monoamines. MAO = monoamine oxidase; MAOI = monoamine oxidase inhibitor; SSRI = selective serotonin reuptake inhibitor; SNRI = selective norepinephrine reuptake inhibitor.



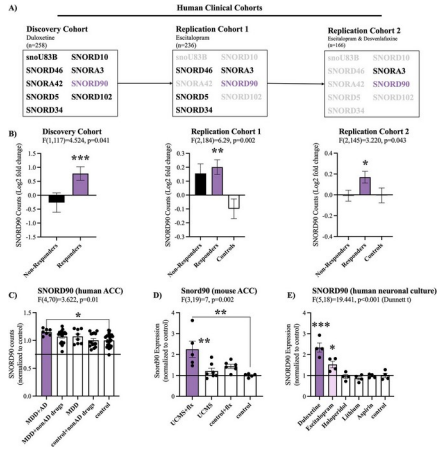
Carlson and Birkett (2022), Fig. 17.14, p.568.

A small RNA changes after antidepressant response

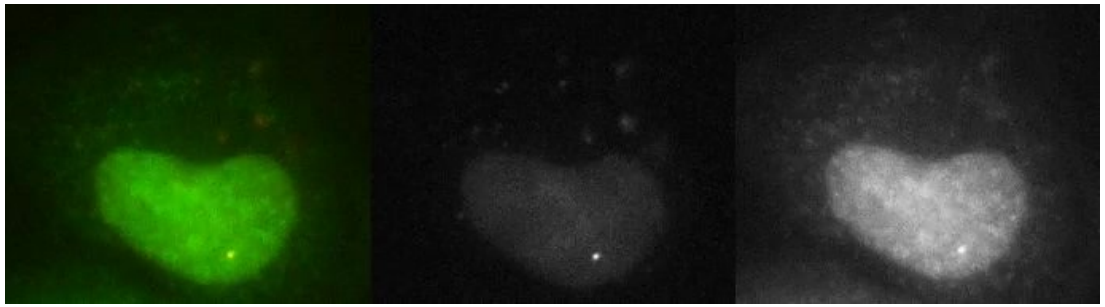
Researchers measured small RNAs in cohorts receiving monoaminergic antidepressants. They identified increased SNORD90 expression after treatment response. A small nucleolar RNA is a regulatory RNA molecule, not a protein receptor through which the medicine directly binds at a synapse.

The investigators then increased Snord90 in the mouse anterior cingulate cortex. This region participates in regulation of behavior and mood-related processing. The manipulation produced antidepressant-like changes in the tested animal behaviors, allowing a causal test beyond the human association.

RNA molecules also move after they are produced. Separate live-cell imaging tracks individual transcripts using different labels for exon and intron sequences. Spliced RNA moves through nucleus and cytoplasm, while the intron signal remains strongest at the transcription site under these imaging conditions.



Lin et al. (2023), Fig. 1.



Spliced RNA moving in nucleus and cytoplasm. Coulon et al. (2014).

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A monoamine deficit is not a universal explanation

The monoamine hypothesis grew partly from the actions of effective medicines and experiments altering transmitter availability. A treatment's action can identify a useful control point without proving that every untreated patient began with a simple shortage of that transmitter.

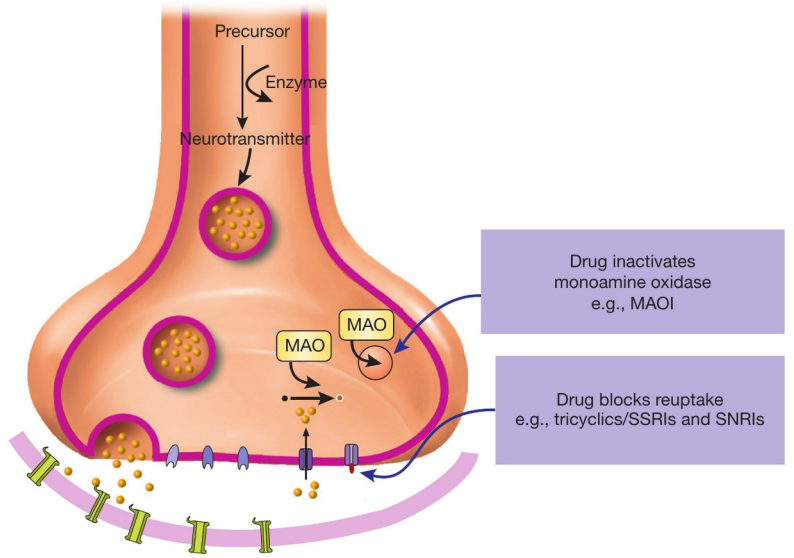
Serotonin acts through several receptor families on different cells. Increasing its availability can therefore change multiple processes at once. The eventual network response depends on receptor distribution, prior activity, and adaptations that occur during treatment.

A mechanistic explanation must connect the immediate drug action to a later functional change. Measurements of synapses, gene regulation, and behavior help trace that sequence. The clinical syndrome remains more heterogeneous than a single chemical concentration measured in one compartment.

er 17

Figure 17.14 Pharmacological Treatment Strategies for Depression

Treatment strategies include decreasing enzymatic deactivation or reuptake of monoamines. MAO = monoamine oxidase; MAOI = monoamine oxidase inhibitor; SSRI = selective serotonin reuptake inhibitor; SNRI = selective norepinephrine reuptake inhibitor.



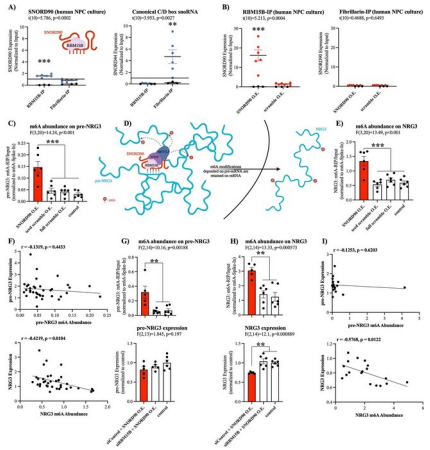
Carlson and Birkett (2022), Fig. 17.14, p.568.

SNORD90 regulates the fate of another RNA

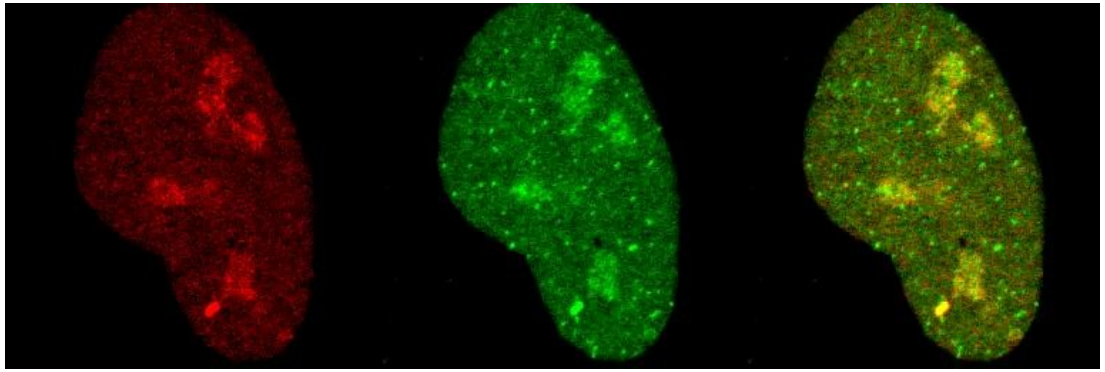
The study identified neuregulin 3, or NRG3, as a target of SNORD90-related regulation. NRG3 participates in neuronal signaling. The investigators traced a pathway in which chemical marking of its RNA changed the likelihood that the transcript would be degraded.

An RNA-binding protein recognized the modified transcript and promoted its decay. Reducing the transcript limited production of the corresponding protein. This supplied a concrete sequence from a regulatory RNA to altered availability of a neuronal signaling molecule.

An intron is an RNA segment normally removed during message processing. Separate imaging detects a small, short-lived population of intron-containing transcripts moving near their production site. This processing step is distinct from SNORD90 changing the survival of NRG3 RNA after transcription.



Lin et al. (2023), Fig. 4.



Short-lived unspliced RNA near its production site. Coulon et al. (2014).

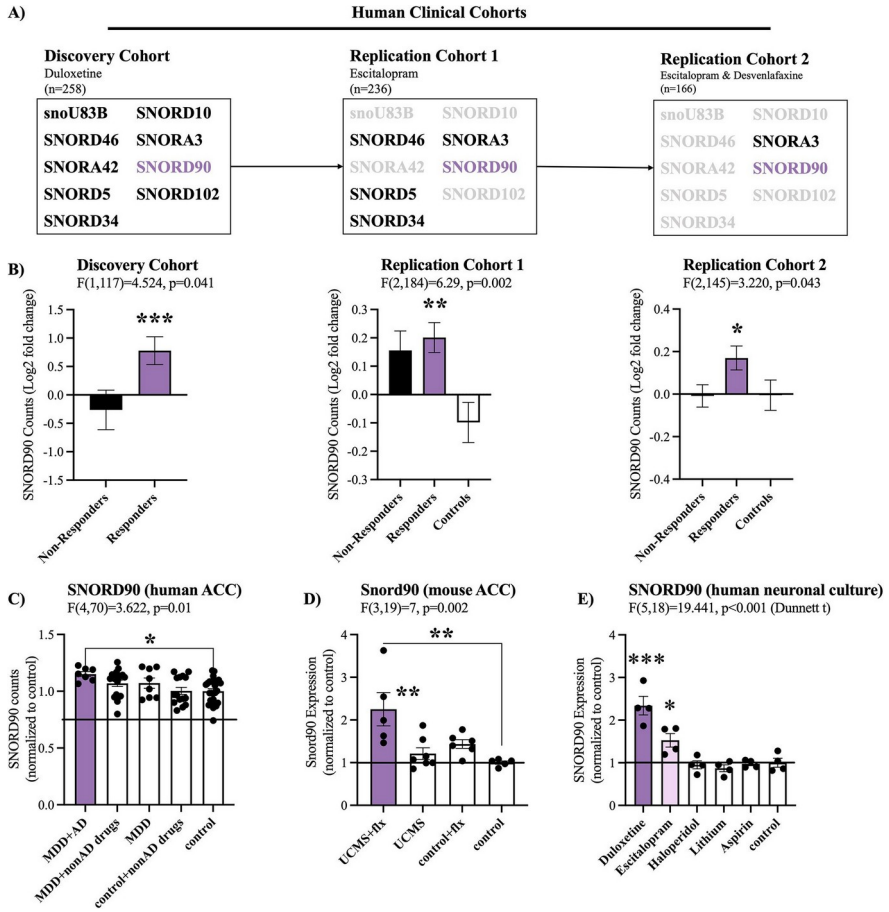
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Neurogenesis is one form of plasticity among several

Neurogenesis is the production of new neurons from precursor cells. Animal studies have examined how stress and antidepressants affect this process in the dentate gyrus. Producing a neuron also requires its survival, maturation, and incorporation into an existing circuit.

Other forms of plasticity change synapses on neurons that are already present. They can alter receptor number, transmitter release, or dendritic structure without producing a new neuron. These mechanisms operate on different timescales and should not be treated as one interchangeable process.

Evidence about adult human hippocampal neurogenesis remains methodologically debated. A rodent manipulation of precursor-cell production does not by itself establish that the same process explains every patient's clinical response. The preparation and the measured cellular stage remain essential.



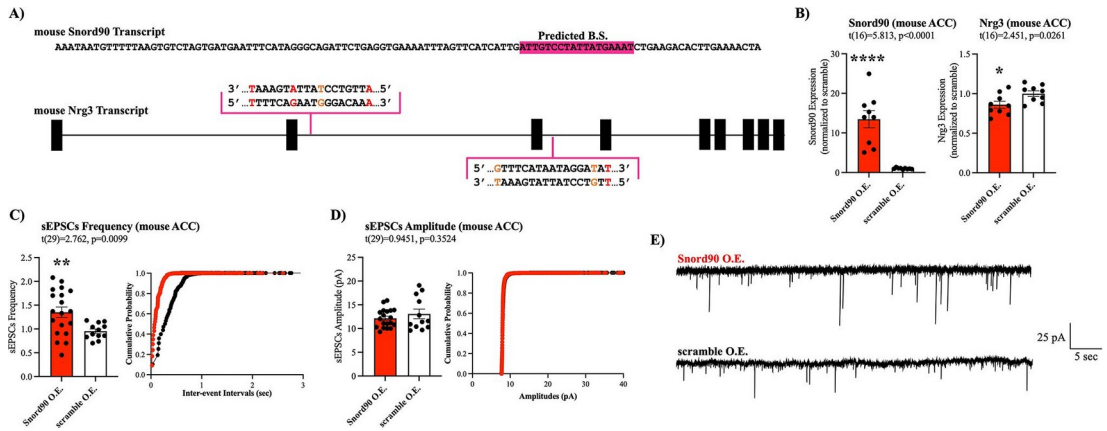
Lin et al. (2023), Fig. 1.

Reduced NRG3 changes cortical glutamate release

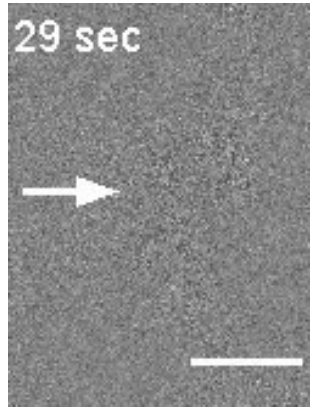
In the mouse anterior cingulate cortex, decreasing NRG3 expression increased glutamatergic release in the study's measurements. Glutamate is the principal excitatory transmitter in many cortical pathways. The RNA-regulatory sequence therefore changed a concrete synaptic output.

This result connected the molecular manipulation to transmitter signaling. Increased Snord90 changed NRG3 RNA handling, reduced NRG3 expression, and altered glutamate release. Each step supplied a testable event between the initial regulatory change and the behavioral phenotype.

Glutamate release can be visualized with a fluorescent extracellular sensor. In a separate cultured-neuron experiment, acute removal of synaptotagmin-1 produces visible spontaneous release events. Each brief fluorescence increase reports local glutamate exposure; the manipulation differs from reducing NRG3 through the antidepressant-associated RNA pathway.



Lin et al. (2023), Fig. 6.



Spontaneous glutamate events after SYT1 removal. Vevea et al. (2020).

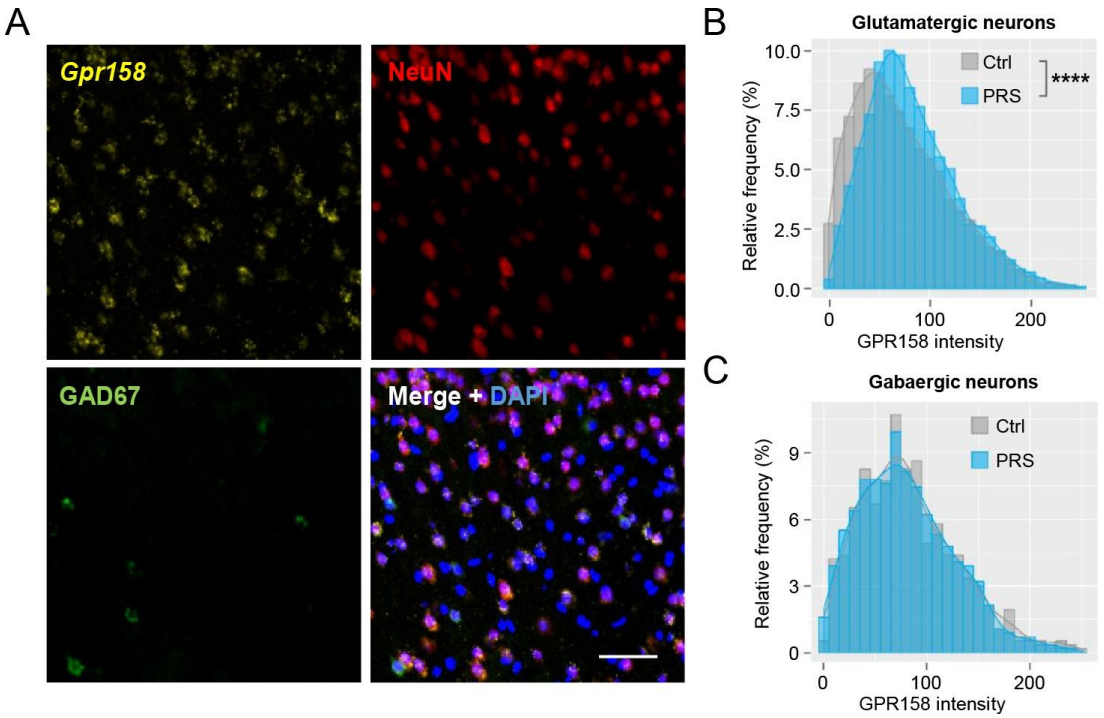
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Chronic stress can change receptor abundance

Glucocorticoids are hormones released through the body's stress-response system. They can alter gene regulation in the brain and other tissues. Repeated exposure can therefore change the amount of a receptor or signaling protein rather than only producing a momentary electrical response.

The prefrontal cortex participates in regulation of behavior, attention, and responses to changing circumstances. Stress-related molecular changes there can influence synaptic function. The affected cell population and the direction of the synaptic change need to be identified experimentally.

Stress is a risk factor with diverse effects, not a synonym for major depression. A stressful experience, a hormone measurement, an animal behavioral response, and a clinical diagnosis describe different levels of observation within a possible causal sequence.



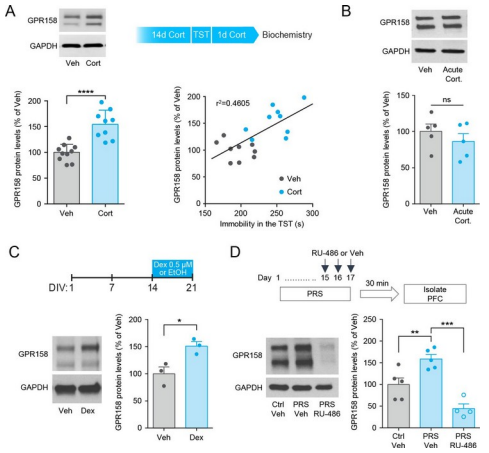
Sutton et al. (2018), Fig. 2.

GPR158 increases after chronic stress in mice

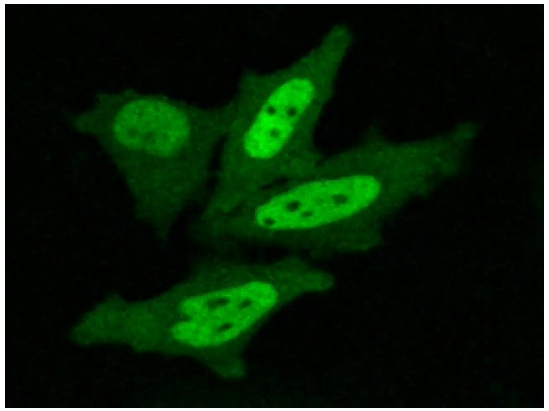
GPR158 is a membrane receptor studied in mood-related biology. Researchers found increased GPR158 in prefrontal samples from people with major depression. They also measured its response to chronic stress in mice, where experimental conditions could be controlled.

Chronic stress increased prefrontal GPR158 in a glucocorticoid-dependent manner. This placed the receptor change downstream from a hormonal response. The experiment therefore examined how stress exposure changed a specific molecular component rather than only comparing two behavioral groups.

Glucocorticoid receptors can move between cytoplasm and nucleus, where they regulate gene activity. A separate living-cell recording follows fluorescent receptor movement during the cell cycle without hormone treatment. It reveals receptor localization rather than the stress-induced GPR158 increase measured in the mouse brain.



Sutton et al. (2018), Fig. 3.



Glucocorticoid receptor movement in living cells. Matthews et al. (2011).

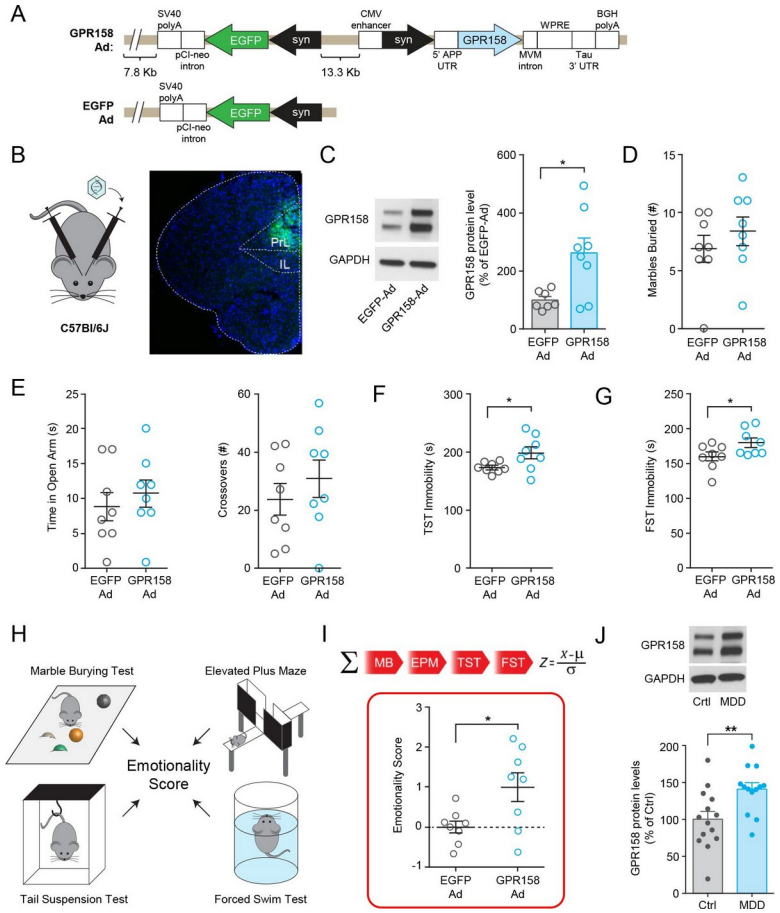
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Increasing the receptor changes mouse behavior

The investigators used a viral method to increase GPR158 expression in mouse prefrontal cortex. This manipulation changed the amount of the receptor in a selected region. They then measured behavior using established tasks relevant to stress and antidepressant research.

Increasing GPR158 produced depression-like behavioral changes in these tests. The term describes the measured animal phenotype and does not mean the mouse reported a human depressive experience. The anatomical manipulation nevertheless allowed a direct test of the receptor's contribution.

The experiment changed a candidate molecular factor rather than simply observing it. It therefore complemented the finding that chronic stress increased GPR158, providing an intervention between receptor abundance and the measured behavioral response.



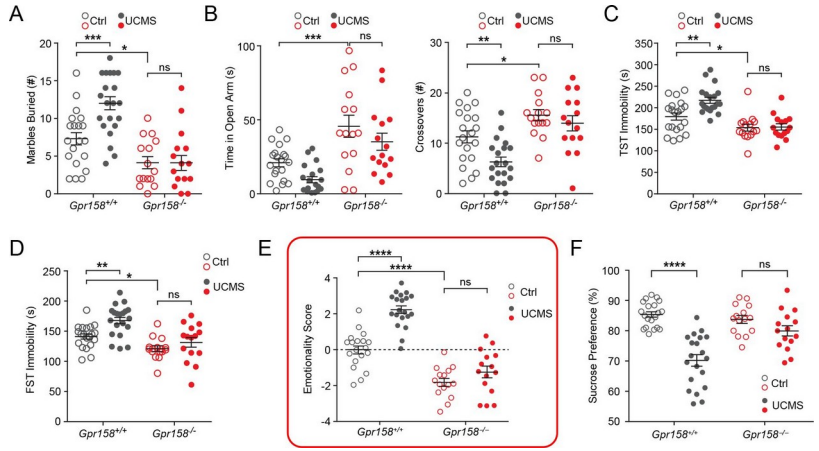
Sutton et al. (2018), Fig. 4.

Stress-related tasks measure specific actions

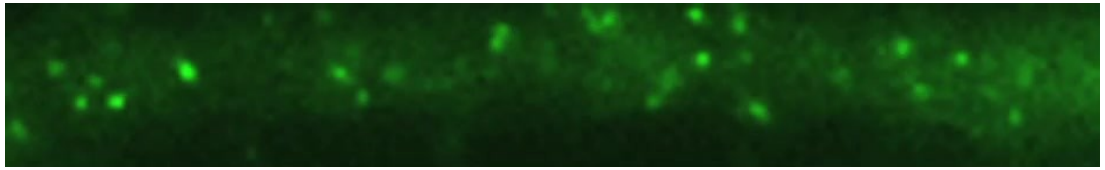
An animal cannot describe persistent sadness, guilt, or subjective hopelessness in the way a human participant can. Researchers therefore measure specific actions such as exploration, reward preference, or responses to controllable challenges. Each task samples only part of mood-related biology.

A change in movement can influence several task scores without changing the same psychological process. Experiments therefore examine relevant behavioral and physiological alternatives. The interpretation should state the action measured rather than assigning an entire clinical syndrome to one result.

Cellular stress effects can be studied separately from animal task performance. A control recording follows vesicles carrying brain-derived neurotrophic factor, or BDNF, along a cultured rat neuron's dendrite. Moving this growth-supporting cargo is a cellular transport process, rather than a measurement of escape behavior or human mood.



Sutton et al. (2018), Fig. 6.



BDNF vesicle transport: untreated dendrite. Adachi et al. (2015).

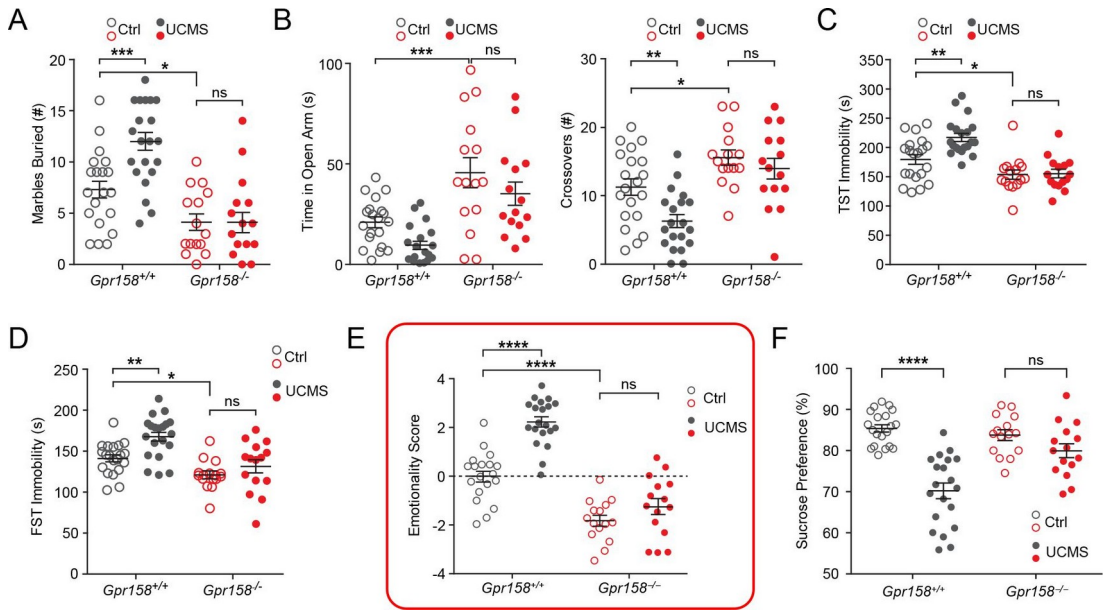
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Removing GPR158 changes resilience to stress

The study also examined mice lacking GPR158. These animals showed an antidepressant-like behavioral profile and greater resilience in the tested stress conditions. Loss and overexpression therefore changed the measured phenotype in opposite experimental directions.

The investigators examined synaptic strength and AMPA-receptor-related function. AMPA receptors respond to glutamate and contribute to fast excitatory transmission. Altered receptor abundance or function can change how strongly a synaptic input influences the postsynaptic neuron.

The pathway thus included a stress-sensitive molecular component and a measurable synaptic effect. The experimental behavioral outcome remained tied to the tasks and preparations used, rather than establishing that removing the receptor would treat every form of human depression.



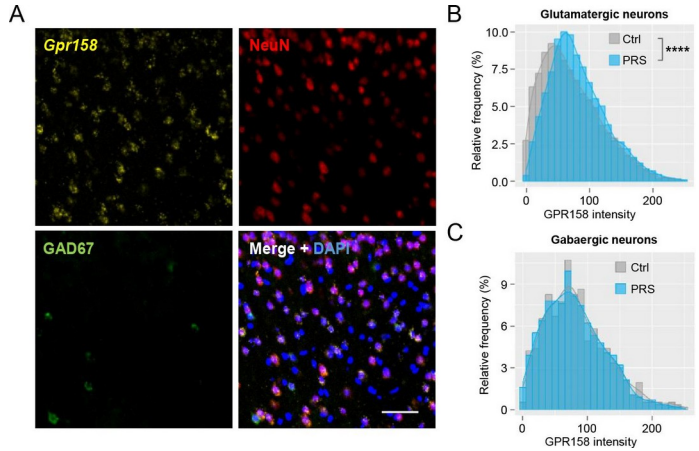
Sutton et al. (2018), Fig. 6.

Hormone-dependent changes can outlast stress

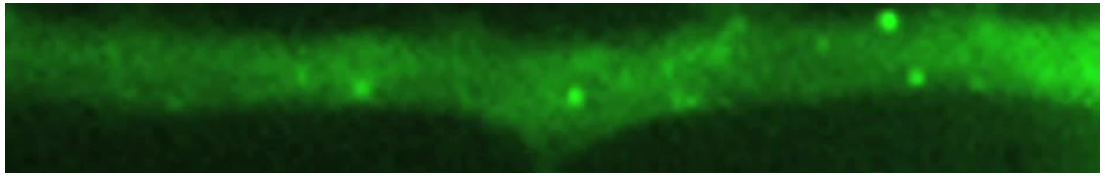
A hormone can initiate a regulatory response that changes how much protein a cell produces. The resulting receptor abundance can outlast the hormone's immediate electrical or metabolic effects. This creates a route through which repeated stress alters later responsiveness.

In the GPR158 study, chronic stress increased the receptor through a glucocorticoid-dependent process. Manipulating the receptor then changed synaptic and behavioral outcomes. The sequence separated an upstream exposure from a downstream cellular component that could be experimentally altered.

Prolonged glucocorticoid exposure can also alter cellular transport. A separate recording follows BDNF-containing vesicles after cultured rat neurons received dexamethasone for twenty-four hours. Its untreated counterpart appears earlier, allowing comparison of cargo movement under defined hormone conditions rather than an inference from behavior alone.



Sutton et al. (2018), Fig. 2.



BDNF vesicle transport after dexamethasone. Adachi et al. (2015).

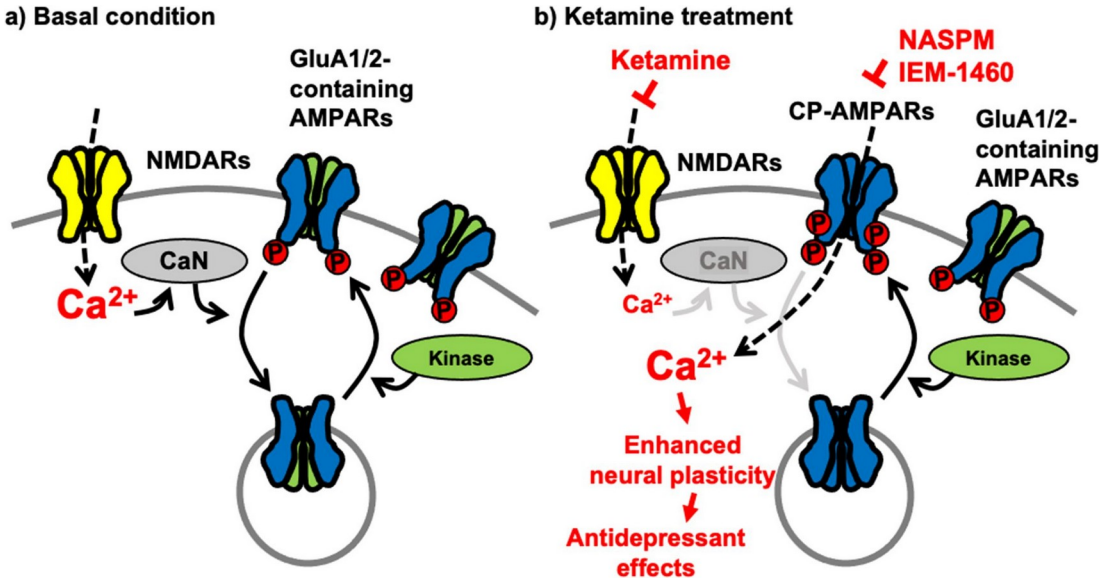
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Ketamine blockade changes downstream pathways

Ketamine is an NMDA-receptor antagonist. NMDA receptors respond to glutamate and can admit calcium under suitable conditions. Blocking them reduces one route of glutamatergic signaling, but the network and intracellular consequences can later change other receptor systems.

The rapid clinical antidepressant effects of carefully administered ketamine-related treatments differ from the slower response common with many monoaminergic medicines. The treatment requires medical supervision because its perceptual, cardiovascular, and other effects vary with dose and clinical circumstances.

A receptor-level description does not fully explain the later response. Researchers examine which cells are affected, which signaling steps change, and how synaptic strength is altered after the initial blockade. These events can occur on different timescales.



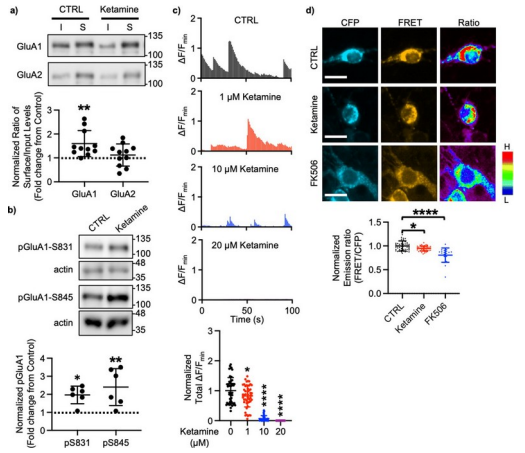
Zaytseva et al. (2023), Fig. 7.

Reduced calcium signaling changes a phosphatase

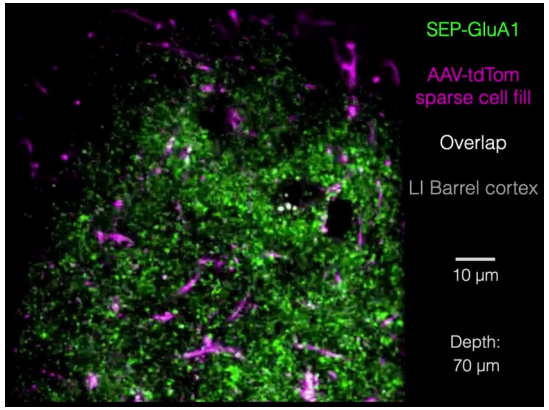
Researchers studied ketamine in cultured mouse hippocampal neurons and in mice. They measured calcium signaling and calcineurin activity. Calcineurin is a phosphatase, an enzyme that removes phosphate groups from selected proteins and thereby changes their functional state.

Ketamine reduced calcium and calcineurin activity in the cultured neurons. With less phosphatase activity, phosphorylation of the GluA1 AMPA-receptor subunit increased. This provided an intermediate step between NMDA-receptor blockade and a change in another glutamate-receptor system.

AMPA receptors can be localized directly at synapses. Separate mouse cortical imaging labels the GluA1 receptor subunit as bright puncta beside filled neuronal branches. These labeled sites identify receptor-containing synapses; this anatomical recording does not show ketamine being applied or phosphatase activity changing.



Zaytseva et al. (2023), Fig. 1.



GluA1-containing synapses across cortical depth. Graves et al. (2021).

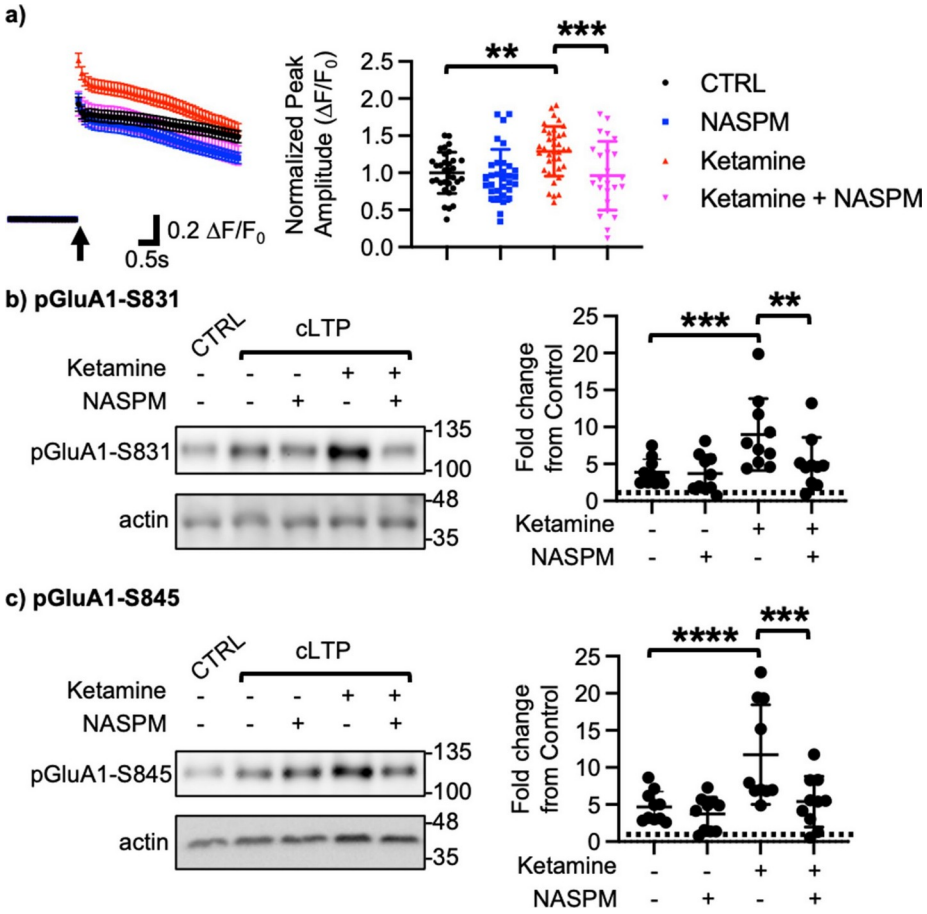
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GluA2 changes AMPA-receptor calcium permeability

AMPA receptors are assembled from protein subunits. Their subunit composition affects their properties. Receptors lacking appropriately configured GluA2 can admit calcium, whereas many GluA2-containing receptors have much lower calcium permeability. The same receptor family therefore contains functionally different channels.

The study found increased expression of calcium-permeable, GluA1-containing AMPA receptors after ketamine treatment. This supplied a route for stronger glutamatergic signaling despite the drug's initial blockade of NMDA receptors. The later receptor population differed from the initial one.

The investigators measured receptor-related and signaling changes in defined preparations. The mechanism concerned a redistribution and modification of existing synaptic machinery rather than creation of an entirely new neural pathway within the treatment interval.



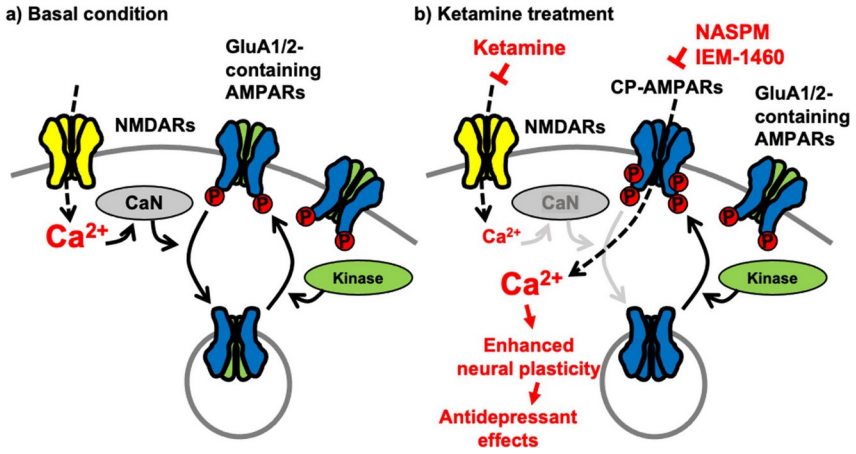
Zaytseva et al. (2023), Fig. 2.

Synaptic strengthening can precede structural remodeling

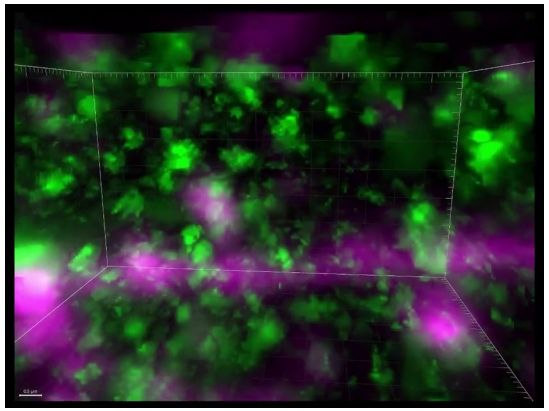
A synapse can become stronger when more receptors respond to transmitter or when existing receptors change function. These changes can occur without immediately adding a new dendritic branch. Electrical efficacy and visible structural growth are different forms of plasticity.

Phosphorylation can change receptor behavior or trafficking. Receptors may also move between intracellular compartments and the surface membrane. The number present at a synapse therefore depends on transport and retention as well as on the rate of protein synthesis.

A separate three-dimensional dendrite recording places GluA1-containing receptor puncta within small protrusions called spines. The shaft contains less of the labeled receptor signal. Existing spines can therefore be examined for receptor localization without assuming that every stronger synapse must have grown a new protrusion.



Zaytseva et al. (2023), Fig. 7.



GluA1 receptor puncta within dendritic spines. Graves et al. (2021).

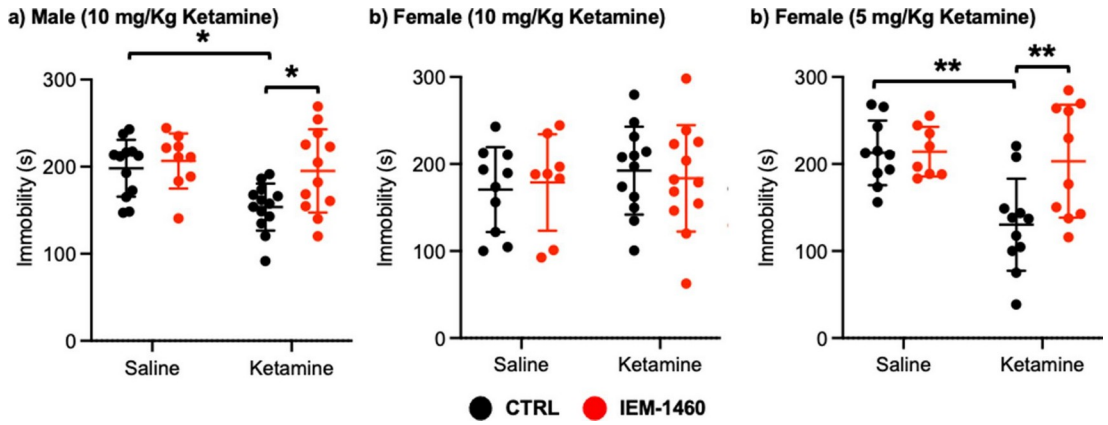
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Receptor blockade removes the mouse response

The investigators tested whether calcium-permeable AMPA receptors contributed to the behavioral response. They administered an antagonist of these receptors and examined ketamine's effects in the animal tasks. This intervention targeted the proposed downstream receptor step.

The antagonist abolished ketamine-associated behavioral effects in the tested conditions. This result supported a necessary role for the receptor population within the experimental pathway. The conclusion concerned the specific preparation, treatment sequence, and behavioral measurements used.

The experiment added causal evidence beyond observing receptor changes after treatment. It interrupted a candidate step and measured whether the later response remained, directly testing whether the synaptic modification participated in the observed outcome.



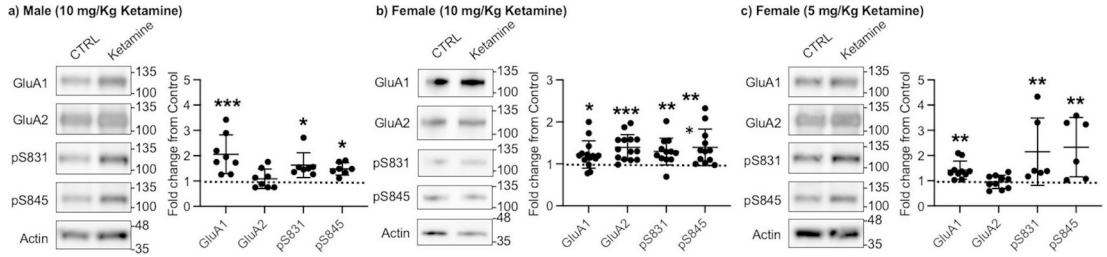
Zaytseva et al. (2023), Fig. 5.

Cells and behavior supply different measurements

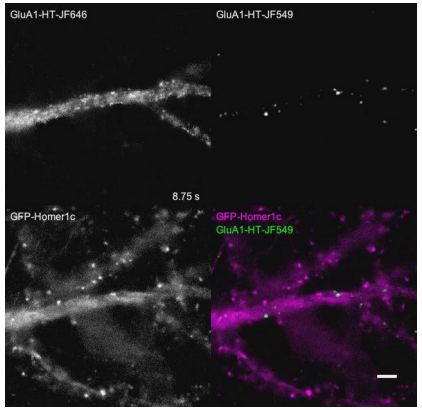
Cultured neurons allowed direct measurements of calcium, enzyme activity, and receptor changes. Whole-animal experiments tested whether related interventions changed measured behavior. The two preparations supplied complementary information, but they did not make every cellular event observable during the behavioral test.

In mice, ketamine increased hippocampal synaptic GluA1 measures within the study's early post-treatment interval. The researchers then examined behavior and receptor antagonism. This placed biochemical and behavioral outcomes within a defined experimental sequence.

Receptor supply can be followed with sequential fluorescent labels. In a separate cultured hippocampal neuron, one color marks previously labeled GluA1 and another marks newly synthesized subunits. Their movement through the dendrite reveals receptor trafficking at a cellular scale, distinct from the mouse behavioral response to ketamine.



Zaytseva et al. (2023), Fig. 3.



Previously labeled and newly synthesized GluA1. Wong et al. (2024).

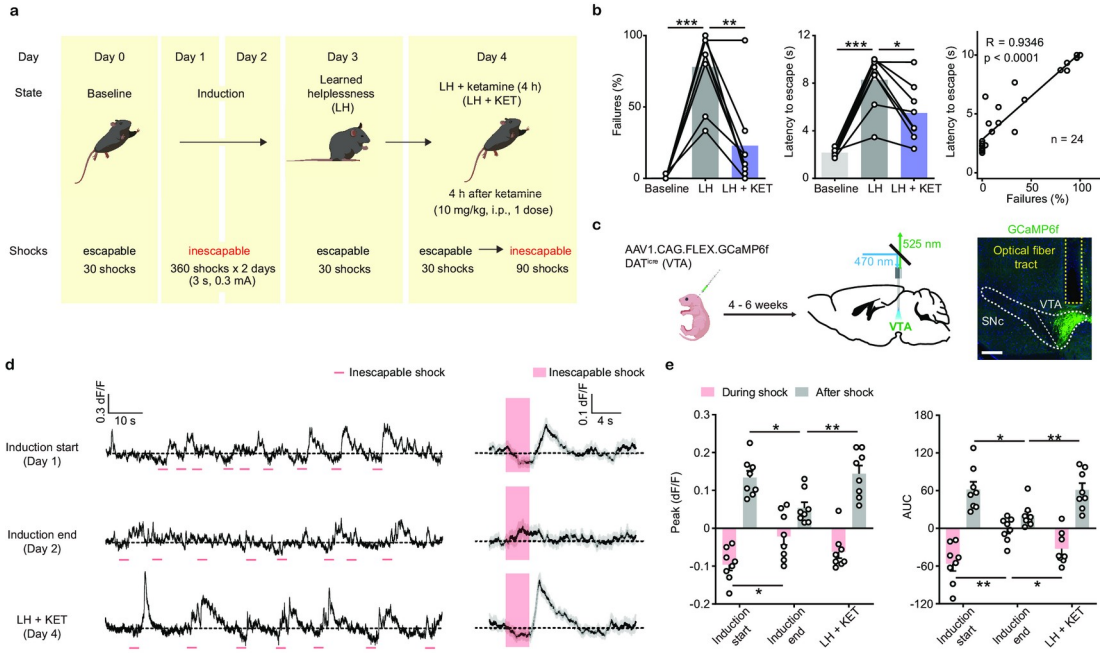
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Learned controllability changes subsequent action

An organism learns whether its actions alter an outcome. When actions consistently fail to change an aversive event, later escape behavior can decrease even when escape becomes possible. Researchers use such tasks to study learning about controllability and responses to stress.

A reduced escape response can arise from several processes, including altered learning, motivation, movement, or perception. Experiments must examine which process is affected. The action measured is whether and how the animal performs the available escape response.

This task captures a defined behavioral adaptation and its underlying circuitry. It does not reproduce every component of major depression. Its value is that investigators can manipulate particular neural pathways while observing a measurable action under controlled conditions.



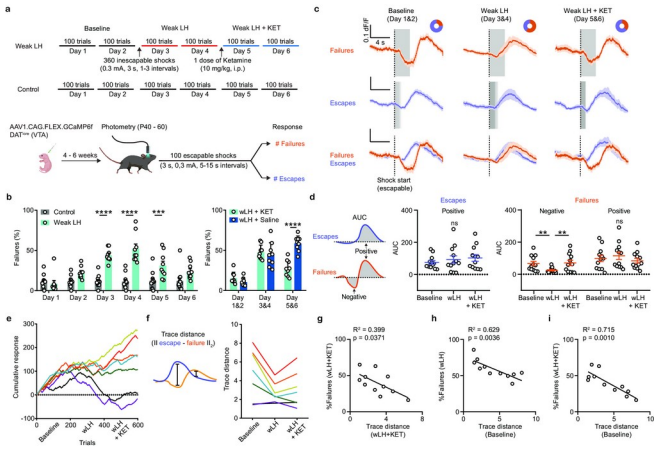
Wu et al. (2021), Fig. 1.

Aversive learning changes dopamine dynamics

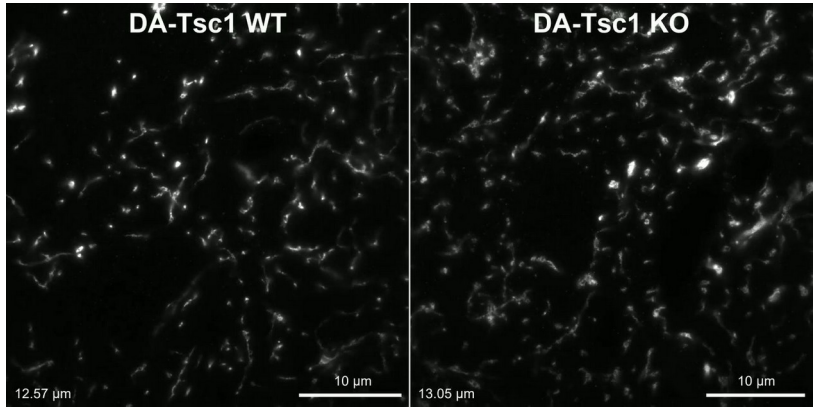
Researchers recorded dopaminergic activity during aversive learning in mice. They examined neurons in the ventral tegmental area and signaling across a prefrontal–midbrain network. The measurements tracked how activity changed as the animals learned about their actions and outcomes.

Dopaminergic dynamics varied across learning and were associated with later sensitivity to ketamine. The result concerned a changing signal during a task, rather than a fixed whole-brain dopamine concentration. Timing relative to behavior was central to the measurement.

Dopamine neurons send axons into the nucleus accumbens. Separate tissue imaging follows labeled branches in mice with or without the growth regulator Tsc1 in dopamine cells. The depth scan depicts pathway anatomy rather than dopamine release during aversive learning or recovery after ketamine.



Wu et al. (2021), Fig. 2.



Dopamine-axon anatomy in nucleus accumbens. Kosillo et al. (2022).

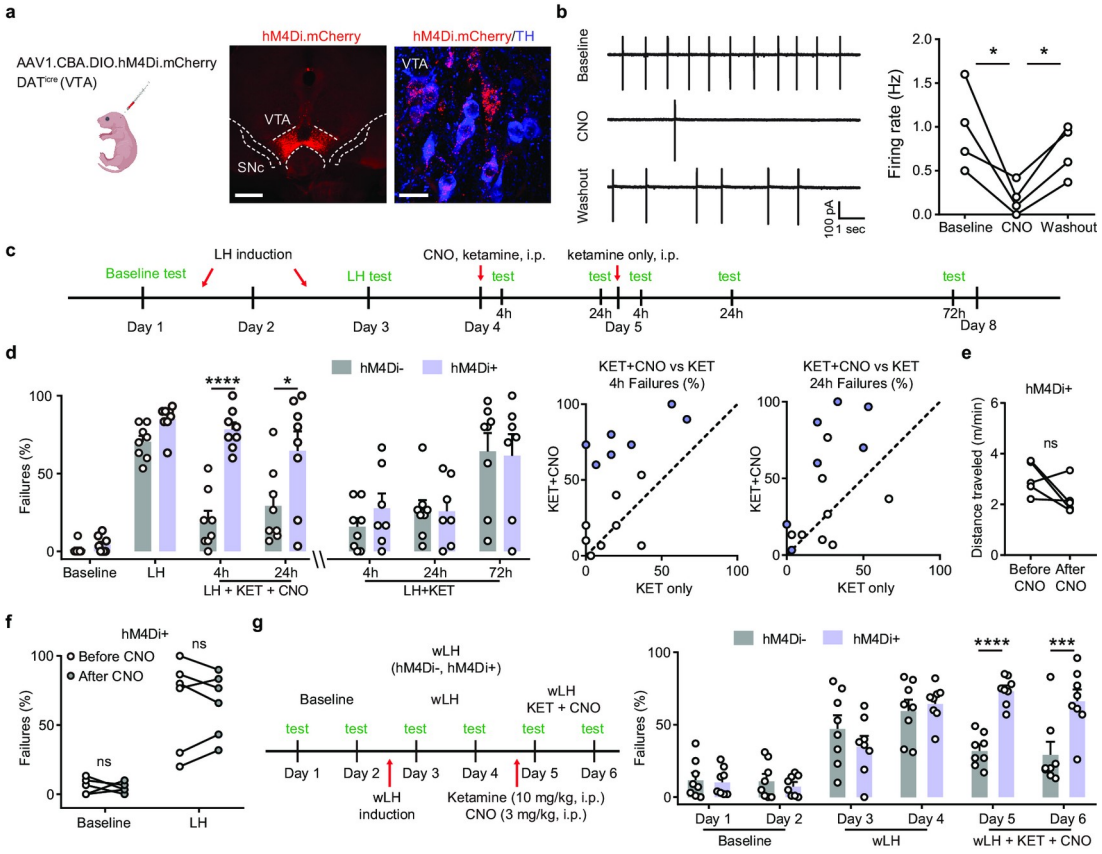
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Inhibiting dopamine signaling blocks behavioral rescue

Chemogenetics uses engineered receptors to change activity in a selected neuronal population when an activating compound is given. The investigators used this approach to inhibit dopamine signaling during their test of ketamine's behavioral effect.

This inhibition blocked ketamine's restoration of escape behavior in the experimental conditions. The result supported a requirement for dopamine signaling within the tested response. It did not mean that ketamine acted only by binding directly to dopamine neurons.

The study examined where ketamine initiated the relevant circuit change. Its results pointed toward actions in medial prefrontal cortex that subsequently altered dopamine dynamics, distinguishing an upstream site of action from a downstream pathway needed for the behavior.



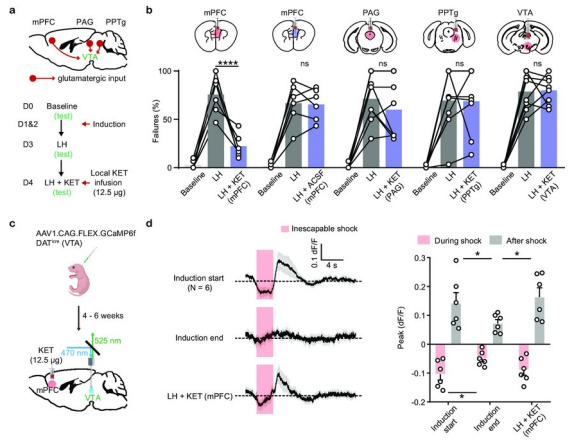
Wu et al. (2021), Fig. 3.

A circuit can regulate another region indirectly

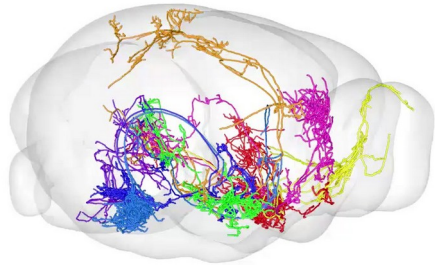
A drug can change a downstream neuron without binding to its main signaling receptor. Cortical neurons project to other regions and recruit local circuits there. Altering cortical activity can therefore change midbrain dopamine signaling through intervening synaptic pathways.

To locate a causal step, researchers can manipulate the proposed upstream cells and measure downstream activity or behavior. Different interventions test sufficiency and necessity. A sufficient manipulation can produce an effect, while a necessary pathway must remain available for a particular effect to occur.

Serotonin supplies another widely distributed modulatory pathway. A reconstruction of six dorsal-raphe serotonin neurons follows their axons through multiple brain regions. These cells arise in a different nucleus and release a different transmitter from the dopamine neurons manipulated in the ketamine experiment.



Wu et al. (2021), Fig. 5.



Six dorsal-raphe serotonin-neuron projections. Ren et al. (2019).

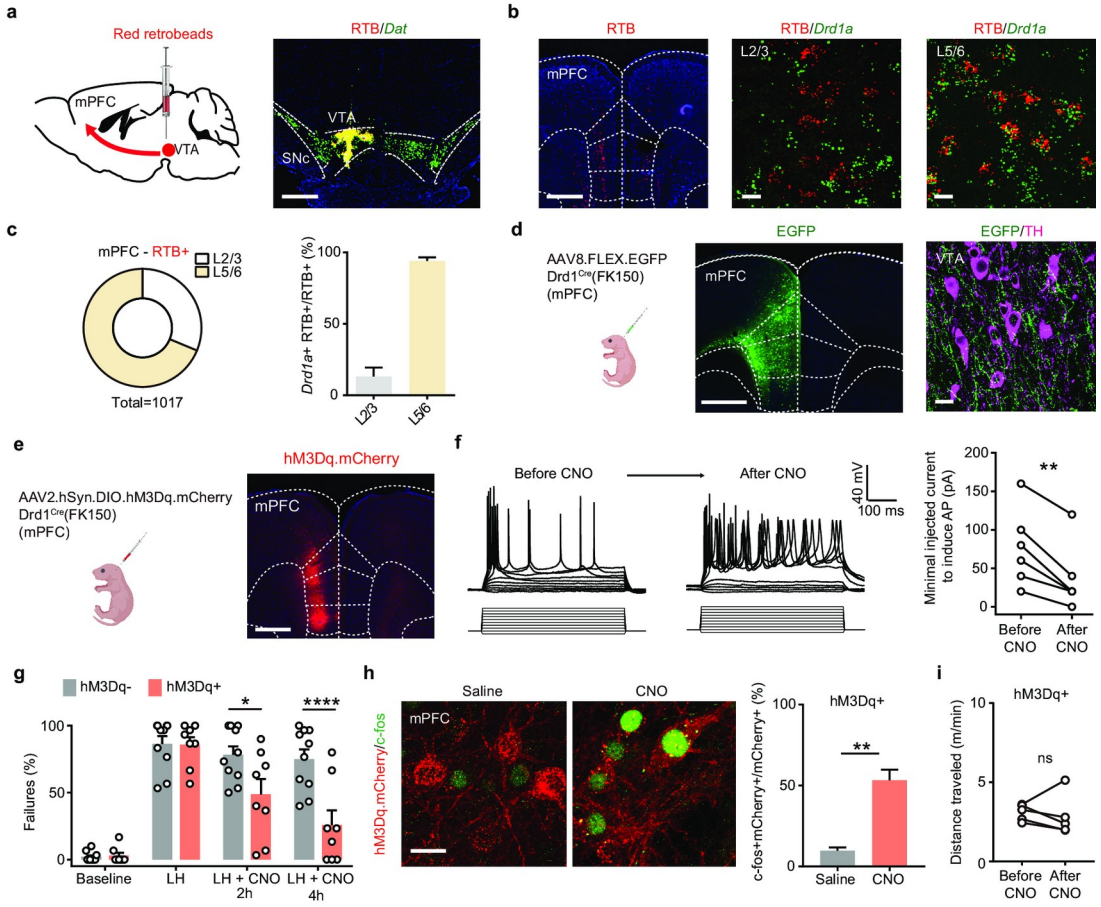
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Activating prefrontal D1 cells mimics the rescue

The researchers activated medial prefrontal neurons expressing the dopamine D1 receptor. This cell population was selected by receptor expression and anatomical location. Chemogenetic activation changed its activity without requiring ketamine to be administered through the same initial mechanism.

Activation reproduced aspects of ketamine's behavioral effect in the escape task. This provided a sufficiency test for the selected prefrontal population. It complemented the necessity test in which dopamine inhibition prevented ketamine-associated rescue.

The two interventions occupied different positions in the circuit. One increased activity in a selected upstream cortical population, while the other reduced dopamine signaling required for the response. Together they specified experimentally tested steps rather than one undifferentiated dopamine explanation.



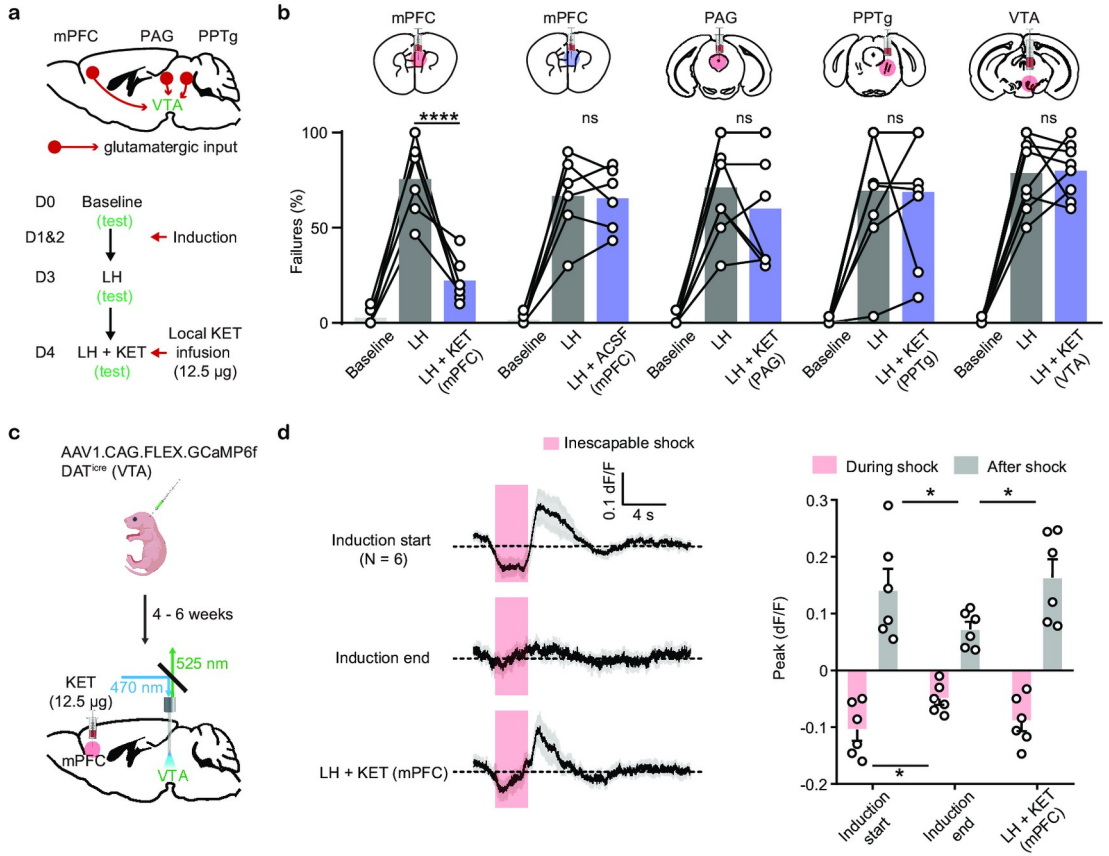
Wu et al. (2021), Fig. 6.

Restored action is measured during a particular task

The measured rescue was renewed escape behavior after the learning procedure. It described an animal performing an available action under defined conditions. The study did not directly measure a verbal report of mood or the full clinical course of human depression.

The physiological recordings located changes in dopamine dynamics across learning and treatment. Their temporal organization mattered because dopamine contributes to action and outcome processing, not simply a uniform level of pleasure. The same transmitter can influence several task-related computations.

The study therefore examined a circuit mechanism for changing a specific learned behavioral state. Translating this mechanism to clinical treatment requires measurements that connect the relevant human circuit changes to symptoms and functioning, rather than assuming that one task exhausts the diagnosis.



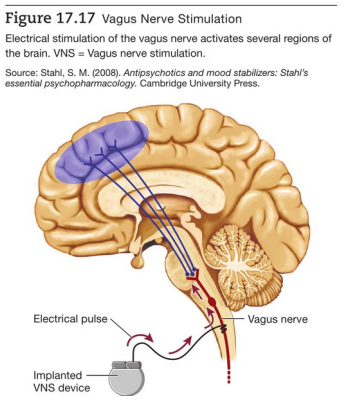
Wu et al. (2021), Fig. 5.

Stimulation treatments target neural networks

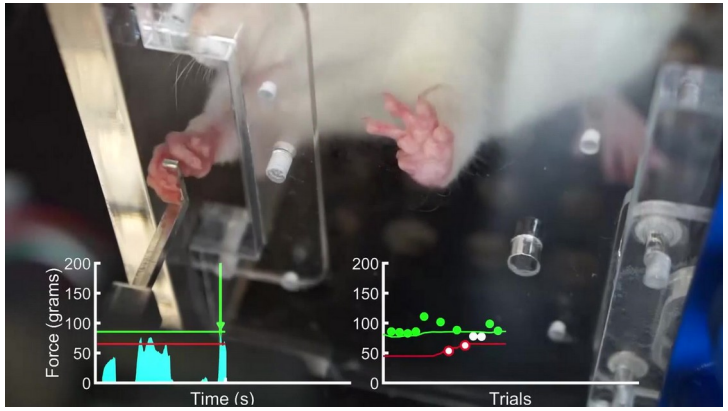
Electroconvulsive therapy is performed under anesthesia and medical monitoring using a controlled electrical stimulus. Repetitive transcranial magnetic stimulation uses changing magnetic fields to influence cortical activity without an implanted electrode. Their procedures and physiological effects differ.

Deep brain stimulation uses implanted electrodes to affect selected pathways and remains a specialized investigational approach for many mood-disorder applications. Choosing a target requires identifying the network the stimulation recruits. A named anatomical site is not itself a complete account of the treatment mechanism.

Vagus-nerve stimulation sends electrical pulses through a peripheral nerve to influence central networks. A separate rat rehabilitation experiment pairs pulses with weaker forelimb pulls after rat spinal injury. Its force trace and stimulation markers describe a timed intervention, rather than a trial of stimulation for human depression.



Carlson and Birkett (2022), Fig. 17.17, p.571.



Timed vagus stimulation during rat rehabilitation. Ganzer et al. (2018).

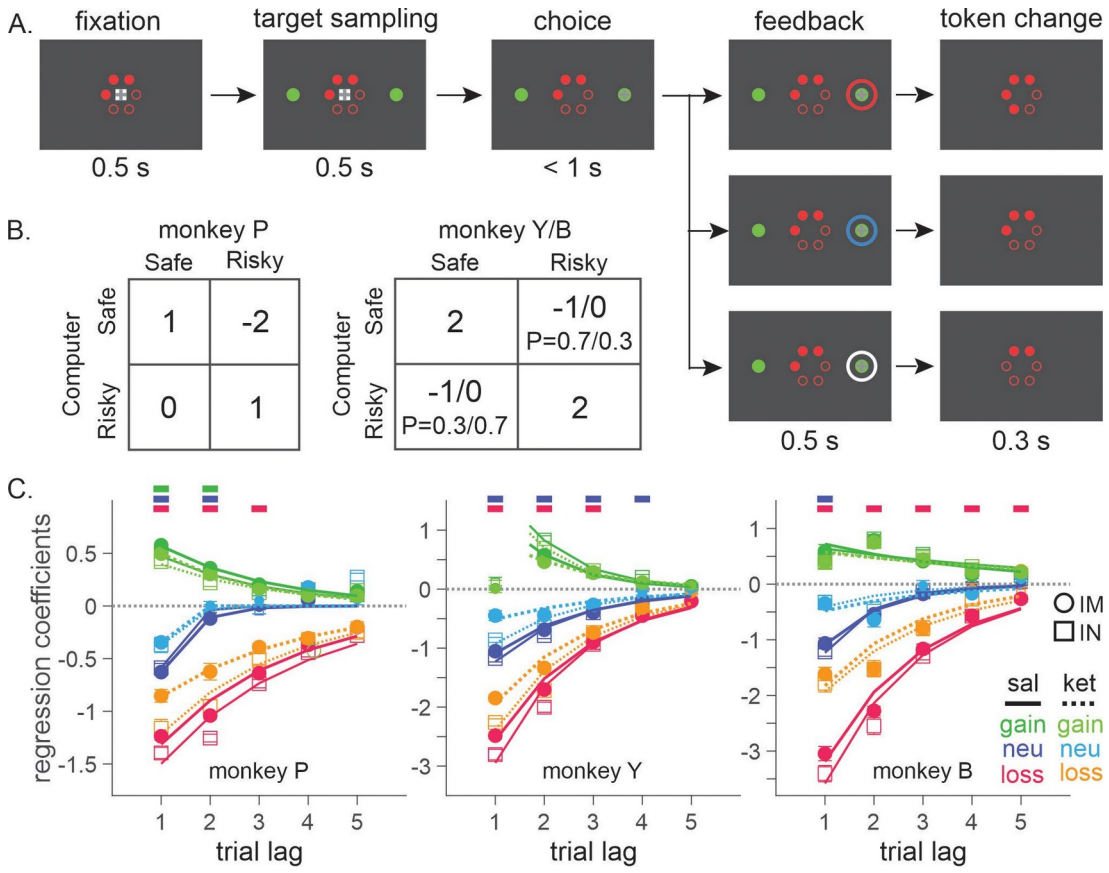
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Ketamine changes how macaques evaluate losses

Researchers studied rhesus macaques making repeated decisions involving gains and losses of tokens. The tokens provided a controlled way to measure how favorable and unfavorable outcomes influenced later choices. Computational models separated several possible components of the decision process.

Therapeutic-range ketamine administration reduced the aversive impact of token losses in the task. The change was more specific than a uniform increase in reward valuation. The investigators separately assessed responses to gains and other aspects of choice and learning.

The experiment measured an acute decision-making effect in macaques. The proposed relevance to antidepressant action concerns how negative events influence subsequent behavior, while the animals' task performance remains distinct from a measured clinical improvement in depressed patients.



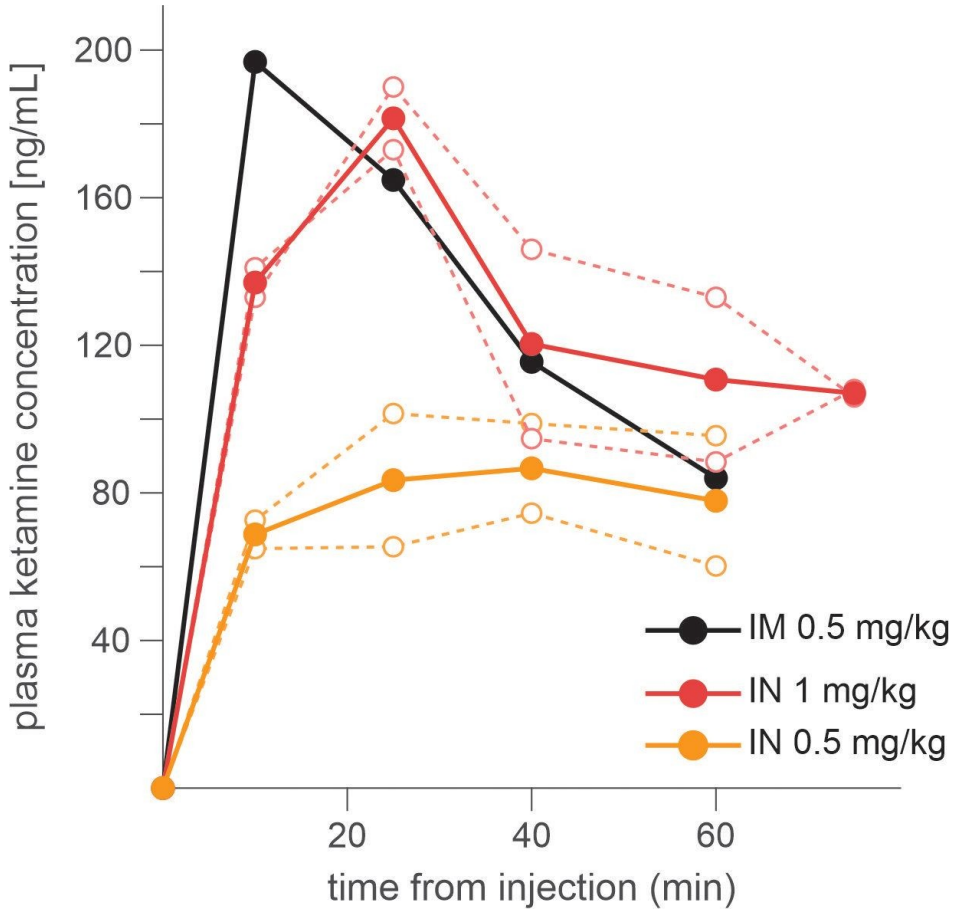
Oemisch and Seo (2024), Fig. 1.

Lithium affects cellular signaling across several pathways

Lithium is used clinically in bipolar disorder, including prevention of recurrent episodes in suitable patients. Its actions involve several intracellular processes rather than a single neurotransmitter transporter. The clinical response and the safest dose require individualized monitoring.

Lithium is eliminated through the kidneys, and its concentration can change with physiology and other medicines. Measuring blood concentrations and relevant organ function helps clinicians maintain an appropriate exposure. A familiar drug can still require close medical supervision.

A proposed molecular mechanism is not equivalent to demonstrated episode prevention. Clinical trials and long-term follow-up establish outcomes in people, while cellular experiments examine particular pathways through which lithium may influence neuronal signaling and plasticity.



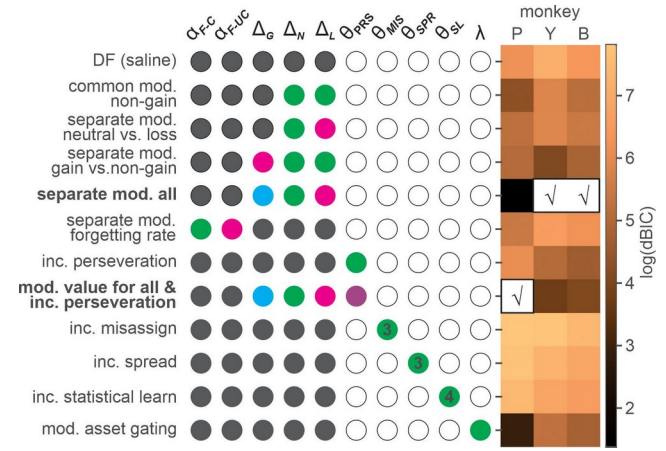
Oemisch and Seo (2024), Fig. 2.

Loss aversion can change selectively

The macaque investigators fitted models that distinguished outcome valuation from perseveration, motivation, and the timing of learning. Perseveration means repeating a choice regardless of newly available evidence. Temporal credit assignment concerns which earlier action receives responsibility for an outcome.

Ketamine reduced the aversive effect of losses without producing comparable changes in every modeled component. This specificity mattered because an animal could otherwise appear less sensitive to loss simply by failing to attend, remember, or perform the task.

A separate rat task presents reward opportunities without punishment risk in the demonstrated block. Captions mark trial events as the animal performs three trials. This baseline condition reveals task execution before adding an aversive possibility, whereas the human study measures choices involving gains and losses.



Oemisch and Seo (2024), Fig. 4.



Rat reward-task trials without punishment risk. Park et al. (2017).

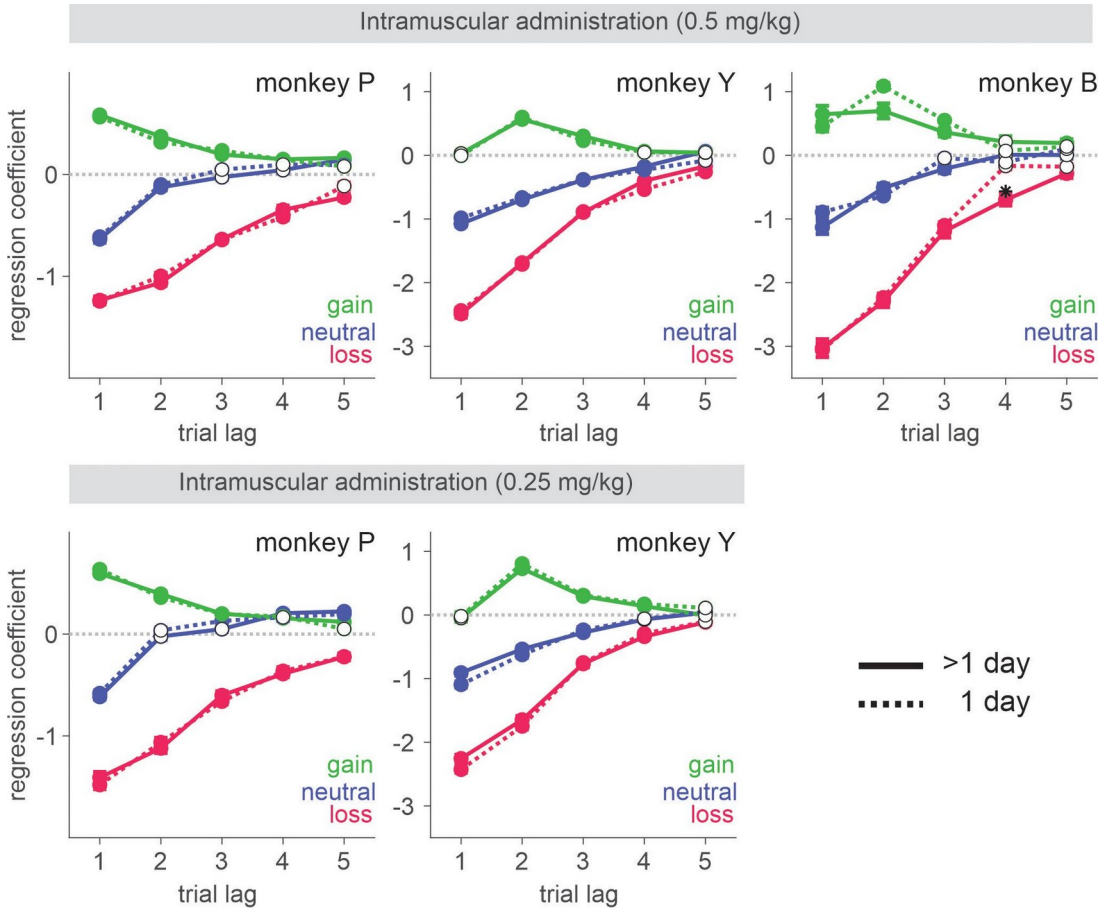
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Clinical response depends on the outcome and its duration

A rapid reduction in symptoms and sustained remission are different outcomes. Remission describes symptoms falling below a defined clinical threshold, while recurrence concerns their later return. A treatment can produce a short-lived change without establishing durable protection against another episode.

Cognition and daily functioning can also recover on a different schedule from mood ratings. Assessment therefore follows more than one dimension over time. The person's goals and adverse effects remain relevant when judging whether an intervention is useful.

Biological treatments are selected within a clinical formulation that includes diagnosis, prior response, medical conditions, and preferences. An educational account of receptor or circuit action does not substitute for that individualized assessment or determine one treatment for every patient.



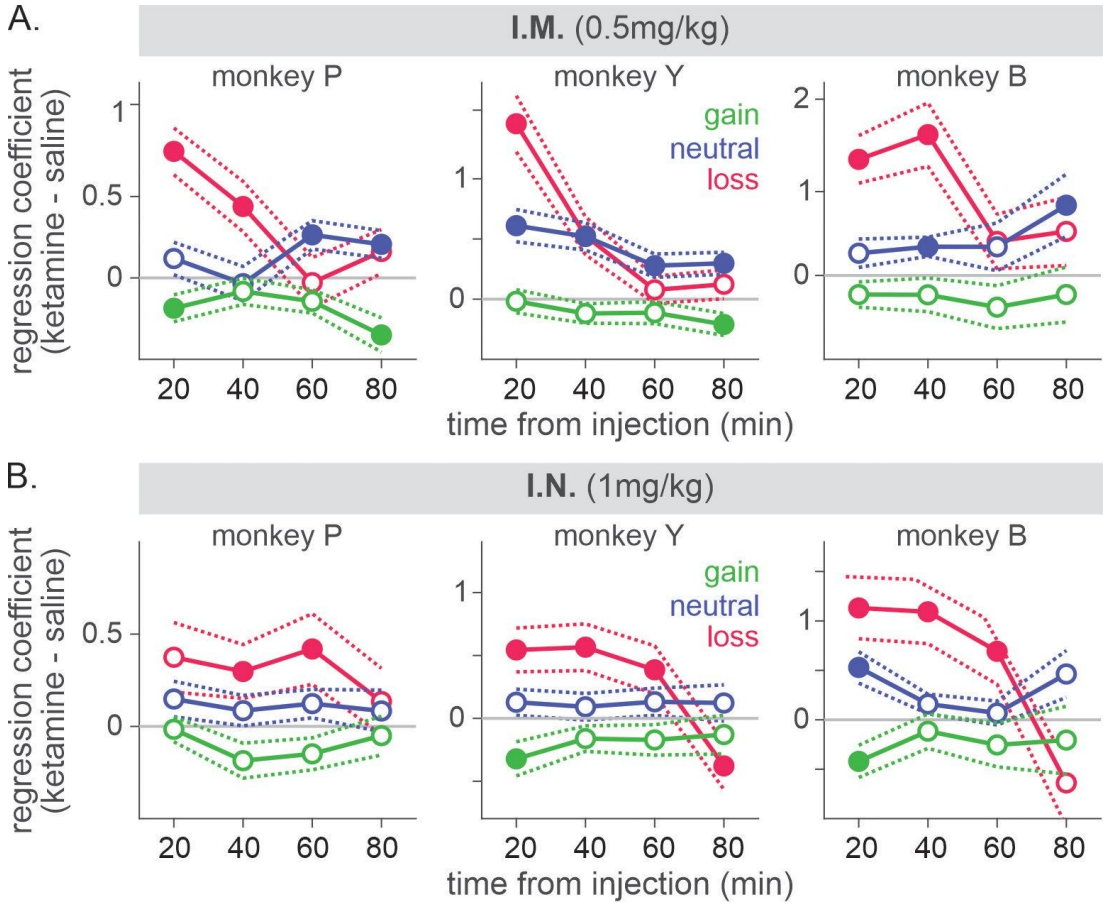
Oemisch and Seo (2024), Fig. 9.

An acute valuation change can alter later experience

A negative event can influence choices beyond the moment it occurs because its memory continues to affect expectations. The macaque study examined models that allowed outcome effects to persist over time. This separated an immediate response from its lingering influence on later decisions.

Reducing the initial impact of a loss could therefore change the information entering subsequent learning. The authors proposed this route as one possible contributor to longer-term antidepressant effects. The experimentally measured change, however, was the acute effect on outcome evaluation.

The distinction identifies what remains to be tested in clinical work. A decision parameter can change during a task without demonstrating that an entire depressive episode has resolved. The measured action and the broader proposed therapeutic mechanism remain separately stated.



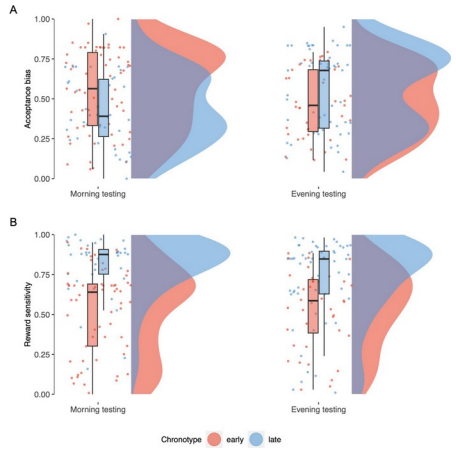
Oemisch and Seo (2024), Fig. 7.

Circadian timing changes when a behavior is measured

A circadian rhythm is an approximately daily cycle generated by biological timing systems. Light helps align these rhythms with the external day. Sleep timing, alertness, hormone release, and other functions can vary across the cycle. The suprachiasmatic nucleus, a small hypothalamic structure, helps coordinate these cycles with information about environmental light.

Chronotype describes a person's tendency toward earlier or later sleep and activity timing. Two people tested at the same clock time can therefore be at different biological phases. A morning measurement does not represent the same internal timing for every participant.

Biological timing also changes animal activity. A recording compares control mice and mice lacking a clock-related regulator early in the dark phase. Their visible movement is measured at the same environmental time, rather than treating an activity difference recorded at an unspecified hour as a stable trait.



Mehrhof and Nord (2025), Fig. 5.



Mouse activity early in the dark phase. Qu et al. (2016).

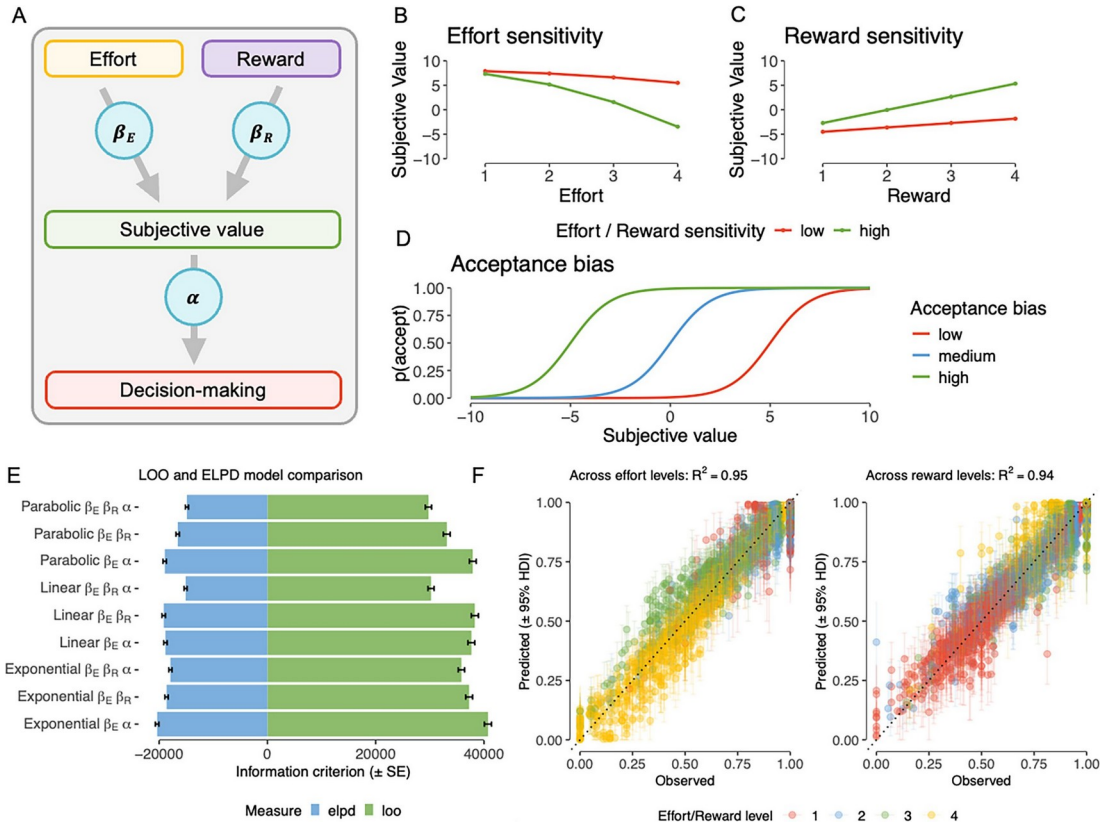
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Effort choices separate reward value from action cost

Researchers measured effort-based decisions using a gamified task in a large participant sample. Participants decided whether to expend effort for outcomes, allowing computational models to estimate how reward and effort contributed to each choice.

Apathy concerns reduced motivation for goal-directed activity, while anhedonia concerns reduced pleasure or reward-related experience. These features can overlap without being identical. The study compared task behavior with both symptom dimensions and diagnostic criteria.

The analysis examined whether altered decisions arose from differences in the value of reward, the cost of effort, or related parameters. This is more precise than treating fewer chosen actions as a direct measurement of one universal motivation system.



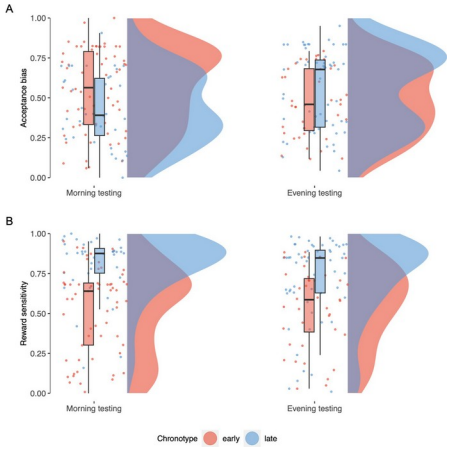
Mehrhof and Nord (2025), Fig. 3.

Testing time interacts with chronotype

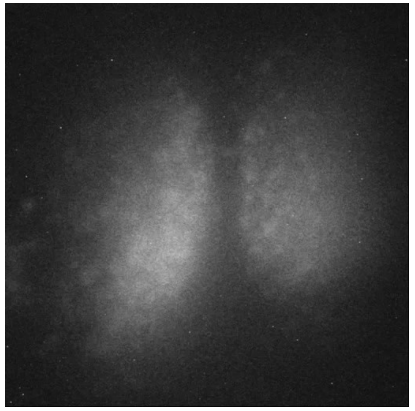
The study compared participants with earlier and later circadian tendencies at morning and evening testing sessions. This directly examined whether the same task produced different decisions depending on the alignment between testing time and the person's timing preference.

Chronotype interacted with time of testing in effort-based decisions. The resulting pattern paralleled aspects of the motivational differences associated with apathy and anhedonia. A task could therefore reflect both clinical features and the participant's biological timing at measurement.

The suprachiasmatic nucleus, or SCN, contains cells that help organize daily rhythms. A separate mouse brain-slice recording uses light produced by a Per2 reporter to follow repeated cycles. Many cells peak together across successive days, providing a cellular timing measurement distinct from a human effort task.



Mehrhof and Nord (2025), Fig. 5.



Synchronized Per2 rhythms in a mouse SCN slice. Hayasaka et al. (2017).

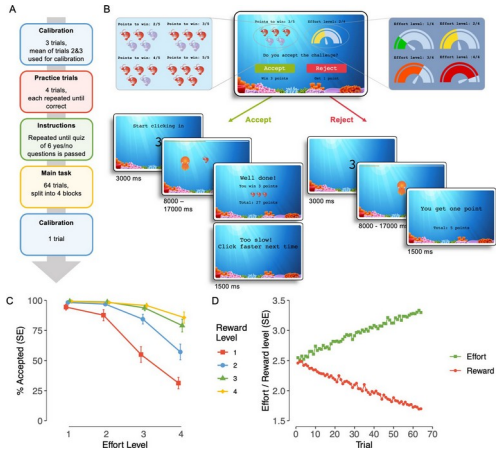
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Timed light changes the phase of the biological clock

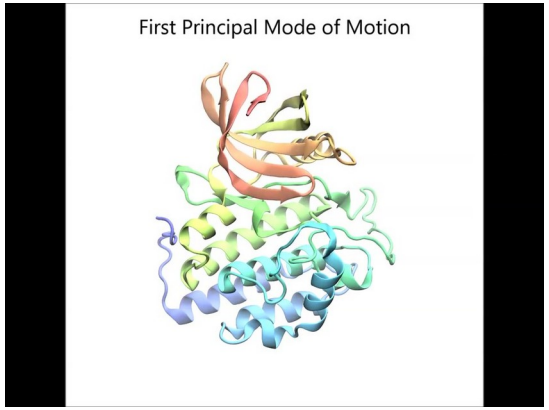
Light reaches circadian control systems through specialized retinal pathways. Its effect depends on when it arrives relative to the internal clock. Light exposure at one phase can advance timing, while exposure at another can delay it. Some retinal ganglion cells contain melanopsin, a light-sensitive protein that helps signal environmental illumination to the clock.

This phase dependence helps explain why the timing of phototherapy matters in seasonal mood symptoms. The goal is not merely a larger total amount of light. Treatment seeks an appropriate change in the alignment between the person's internal rhythms, sleep, and the external day.

Clock timing also depends on proteins that regulate the stability of PERIOD proteins. A published structural model of casein kinase 1 delta shows bending and twisting motions between enzyme regions. These are calculated molecular motions, rather than a recording of light entering the eye or resetting an intact brain.



Mehrhof and Nord (2025), Fig. 2.



Calculated casein kinase 1 delta motions. Philpott et al. (2020).

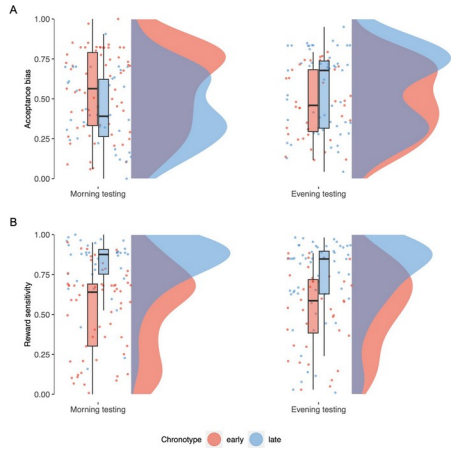
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Circadian timing can imitate a symptom-related difference

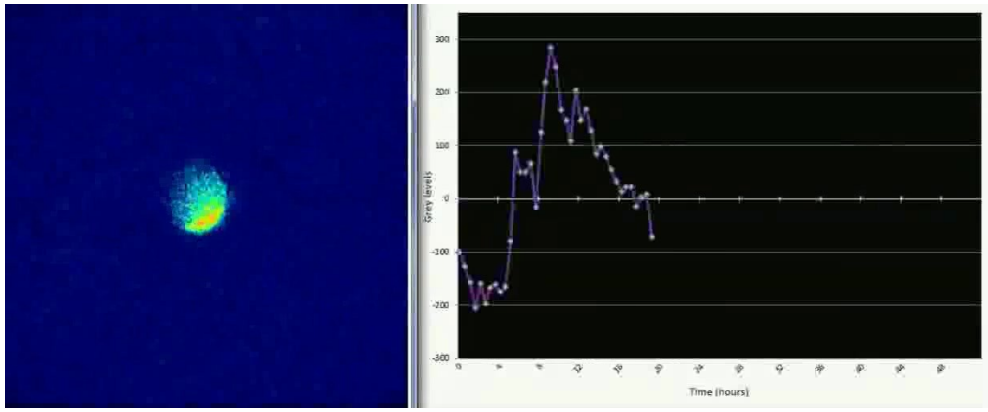
A participant tested at a poorly aligned biological time may value effort differently from the same participant tested at a better aligned time. If testing schedules differ between groups, that variation can contribute to an apparent diagnostic difference.

The study demonstrated a measurable chronotype-by-time interaction rather than simply recommending that researchers record sleep. Its task and modeling approach showed which behavioral computations changed with timing, allowing comparison with symptom-related differences in the same decision framework.

Daily molecular rhythms can also be recorded within an individual cultured cell. A fluorescent reporter rises and falls across forty-eight hours in this separate preparation. Its changing expression provides a cellular time series, while the human study measures how testing time influences motivation-related behavior.



Mehrhof and Nord (2025), Fig. 5.



Rhythmic reporter expression over forty-eight hours. Torres et al. (2016).

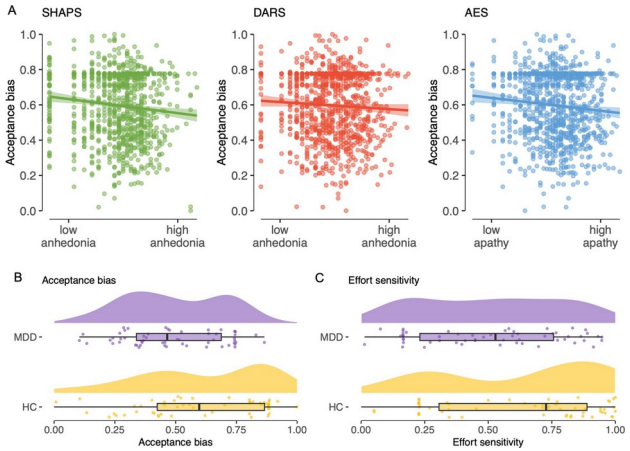
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A behavioral marker needs a defined physiological context

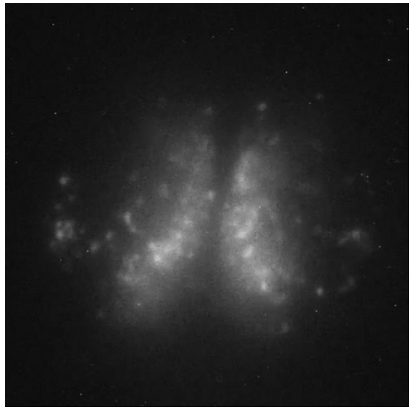
Computational psychiatry uses models to describe processes contributing to behavior. A model parameter is an estimate based on a particular task and dataset. Its clinical interpretation depends on whether relevant nonclinical influences have been measured and separated.

In the effort study, circadian tendency and testing time changed decisions in ways relevant to motivational symptoms. Recording those conditions made the parameter more interpretable. A numerical estimate without that context could combine several biological processes into one apparent deficit.

Cellular synchrony is one part of a stable biological clock. In a separate SCN slice lacking Sik3, individual Per2 reporter rhythms become desynchronized. The cells' differing phases are visible over several days, providing a defined physiological disturbance rather than a direct measurement of apathy or anhedonia.



Mehrhof and Nord (2025), Fig. 4.



Desynchronized Per2 rhythms in a Sik3-null SCN. Hayasaka et al. (2017).

Play video

Key takeaways

Symptoms describe distinct functions. Positive, negative, cognitive, depressive, and manic symptoms vary within and across diagnoses; psychosis alone does not identify one underlying disorder.

Risk and mechanism are different claims. Genetic and environmental associations identify probabilities, while causal experiments must specify the altered cell, molecule, pathway, and measured response.

Circuit coordination depends on recruited inhibition. Pyramidal-to-interneuron communication, gamma timing, and thalamocortical sleep events describe different forms of population organization.

A drug has several biological timescales. Receptor binding, enzyme regulation, RNA stability, synaptic receptor trafficking, behavior, and sustained clinical response need separate measurements.

Animal actions remain defined experimental outcomes. Escape, exploration, or effort choices can test specific pathways without reproducing every subjective and functional feature of human depression.

Biological timing changes interpretation. Chronotype and testing time can alter motivation-related measurements; timed light and other treatments act on particular physiological targets within individualized care.

Carlson & Birkett (2022); primary studies in notes